

Aharonov–Bohm Signature of a Field Response

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The Aharonov–Bohm effect is conventionally understood as a gauge phenomenon requiring a charged probe, leaving neutral particles blind to an enclosed field. However, confined fields carry energy-momentum that warps exterior spacetime, imprinting a geometric holonomy. Gauge holonomy and spacetime warping couple within a composite configuration (A_θ, K_k) on the coaxial cylinder enclosing the solenoid. In the exact flat Einstein–Yang–Mills Cauchy development, the geometric deformation acts as a physical spacetime contraction rate governing photon optical transit times. While a charged probe reads the gauge holonomy θ (masking its metric phase by 10^{35}) and a neutral probe isolates the metric warping k along geodesics, the field configuration determines the response operator that reads it, resolving the Aharonov–Bohm distinction as a joint gauge-gravitational phenomenon.

The field’s gauge-reduced curvature Hessian acts on coupled fluctuations carrying the tensor-product representation of both sectors. In cylindrical coordinates, the connection shifts orbital angular momentum modes, and the regularized heat trace evaluates to a winding sum of joint gauge-spacetime characters. This establishes a strictly positive, stable spectral gap between locally gauge-equivalent configurations, distinguishing them without an external test particle and preserving the spectral separation throughout the spacetime development.

The field’s gravitational curvature is accessible on a tabletop via a dual-arm optical lattice atom gradiometer using laser-cooled, spin-zero ^{88}Sr atoms. Because the ground state has zero magnetic dipole moment, first-order Zeeman shifts and magnetic forces vanish. Modulating the solenoid current at frequency ω forces the quadratic stress-energy tensor to oscillate at 2ω . Differential interrogation across both arms cancels Earth’s uniform gravitational background, while lock-in detection rejects the static apparatus rest mass by 100 dB. In a 40 T configuration with a ten-second lattice hold, the differential metric phase reaches $0.29\ \mu\text{rad}$, permitting five-sigma detection within four days of averaging.

I. LOCAL CURVATURE VS. GAUGE HOLONOMY

The Aharonov–Bohm effect established that vanishing local field strength does not exhaust the physical content of a gauge field [1]. In its canonical formulation, coherent matter-wave packets of charged particles (such as electrons) traverse paths on opposite sides of a confined magnetic flux cylinder. Outside the cylinder, the electromagnetic field tensor vanishes identically, yet the recombined interference pattern reveals the gauge-invariant nonintegrable phase factor [2, 3]

$$\rho_q(\text{Hol}_A(\gamma)) = \exp\left(-\frac{iq}{\hbar} \oint_\gamma A\right). \quad (1)$$

In the modern geometric formulation of quantum phases, Simon [4] recognized that Berry’s adiabatic phase [5] is mathematically the holonomy of a connection on a Hermitean line bundle over parameter space, a framework subsequently extended to non-Abelian holonomies by Wilczek and Zee [6] and non-adiabatic cyclic paths by Aharonov and Anandan [7]. Because the phase shift in Eq. (1) is proportional to the electric charge q , the canonical literature concluded that the Aharonov–Bohm

effect is an exclusively electromagnetic gauge potential phenomenon, and neutral particles—specifically photons—cannot detect the enclosed field because their phase factor is identically unity.

A. Gravitational back-reaction of confined fields

This conventional interpretation rests on the idealization that the background spacetime outside the flux is inert, flat Minkowski space. In reality, all physical particles—charged electrons as well as neutral photons and atoms—propagate through physical spacetime along geodesics governed by the metric $\mathbf{g}_{\mu\nu}$. An electron with mass m_e and charge $q = -e$ acquires both an electromagnetic phase and a gravitational proper-time phase,

$$\Delta\phi_e^{\text{total}} = \underbrace{-\frac{e}{\hbar} \oint_\gamma A}_{\text{Gauge phase}} + \underbrace{\frac{m_e c^2}{\hbar} \oint_\gamma d\tau}_{\text{Spacetime geodesic phase}}. \quad (2)$$

In general relativity and geometric field theories, an electromagnetic field possesses non-vanishing energy density, pressure, and stress. A confined flux cylinder carries mass-energy that inevitably *warps the spacetime geometry around it* via Einstein’s field equations $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}^{\text{EM}}$.

Historically, this spacetime deformation went unexamined in Aharonov–Bohm experiments because the ex-

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treme charge-to-mass hierarchy of the electron establishes an electromagnetic-to-gravitational coupling ratio $\Delta\phi_{\text{gauge}}/\Delta\phi_{\text{grav}} \sim (e/m_e)/\sqrt{G} \sim 10^{35}$. In electron beam interferometry [3], the gauge phase shift is macroscopic ($\sim 10^3$ – 10^5 rad) while the metric geodesic shift from the enclosed field is minuscule ($\sim 10^{-32}$ rad). This practical numerical dominance was mistaken for an absolute physical principle, obscuring the metric deformation induced by the enclosed energy.

Even when electromagnetic shielding completely eliminates exterior fields, the surrounding spacetime carries a metric conical defect, an extrinsic curvature imprint, and a non-trivial spacetime holonomy. Because neutral probes experience zero gauge phase, they bypass the 10^{35} electromagnetic masking, revealing the spacetime metric warping induced by the confined field.

B. Joint Gauge-Metric Holonomies

Gauge holonomy and spacetime warping emerge not as disjoint phenomena coupled post hoc, but as orthogonal sectors of a composite field configuration $\Psi = (\nabla^N, K)$ [8]. The configuration combines an internal metric connection ∇^N and a symmetric tensorial shape field K into a single composite metric connection $\nabla_\Psi^\oplus$ on the Whitney sum bundle $TM \oplus N$. Such composite geometric connections have classical roots in Kaluza–Klein theory [9] and MacDowell–Mansouri formulations of gravity [10], closely connecting to the broader program of formulating general relativity and gauge theories via loop holonomies and connection variables [11, 12]. Here, $\nabla_\Psi^\oplus$ defines the total geometric structure whose squared curvature determines the curvature energy functional $\Phi(\Psi)$ and its gradient response $\mathcal{V}_\Psi$.

At a curvature-flat critical configuration Ψ (where all local curvatures vanish, $\mathcal{F}_\Psi^\oplus = 0$), the second variation of Φ defines a gauge-reduced Hessian operator L_Ψ . On the coaxial cylindrical geometry $M = [R_1, R_2] \times S^1 \times [0, L]$ surrounding the solenoid, the aligned family is defined by

$$A_\theta = \theta d\phi J, \quad K_k = k(d\phi)^2 e_1, \quad k \neq 0, \quad (3)$$

where $\theta \in \mathbb{R}/\mathbb{Z}$ fixes the $U(1)$ gauge holonomy, $\phi \in [0, 2\pi)$ is the azimuthal coordinate encircling the flux core, and $k \in \mathbb{R} \setminus \{0\}$ parameterizes the amplitude of the spacetime shape warping. On every contractible open set $U \subset M$, configurations with identical k and different θ are strictly gauge equivalent via a single-valued internal gauge transformation. Locally, all field strengths and curvature invariants vanish identically.

The field’s own gauge-reduced Hessian $L_{\theta,k}$ distinguishes these locally identical configurations through its spectral response, acting as an intrinsic probe without requiring an external particle.

Joint bundle holonomy decomposition.—The total holonomy of $\nabla_{\Psi_{\theta,k}}^\oplus$ around the azimuthal circle encircling the flux decomposes into commuting blocks,

$$H^\oplus(\theta, k) = H^g(\theta) \oplus H^K(k) \oplus I. \quad (4)$$

A charged matter probe (electron) couples to the gauge block $H^g(\theta)$ (with its metric geodesic phase suppressed by 10^{35}), yielding the character $\cos(2\pi m\theta)$; a neutral probe (photon or atom) couples to the spacetime metric warped by K_k , isolating the geometric holonomy $H^K(k)$ with character $\cos(2\pi mk)$.

Tensor-product field response.—The field’s own fluctuation modes carry the *tensor-product representation* $H^g \otimes H^K$. In cylindrical coordinates, the gauge potential shifts the orbital angular momentum $m \rightarrow m - \theta$, governing radial Bessel modes of shifted order $\nu = |m - \theta|$. Through the Poisson summation and winding expansion, the regularized heat trace contains the product

$$\cos(2\pi m\theta) \cos(2\pi mk). \quad (5)$$

Certified spectral gap.—Subject to coaxial boundary conditions at the solenoid shield $r = R_1$ and outer wall $r = R_2$, the radial eigenvalues are strictly monotonic in $|m - \theta|$ by Watson’s theorem. This monotonicity generates a strictly positive, stable spectral gap between the heat traces of locally gauge-equivalent configurations,

$$\Xi_{1/2,k}^{\text{ch}}(t) - \Xi_{0,k}^{\text{ch}}(t) > 0, \quad (6)$$

which is proven stable across an open neighborhood of spectral weight functions. In contrast to classical spectral geometry on flat bundles where one studies the Laplacian of a fixed external connection [13–17], the operator here is the second-variation Hessian of the field theory itself, dynamically coupling gauge and metric fluctuations.

Spacetime development and photon geodesics.—These configurations satisfy the vacuum Einstein–Yang–Mills constraints and generate exact flat Lorentzian spacetimes $(M_k, \mathbf{g}_k)$ where k is the metric contraction rate. Photon null geodesics measure k via optical transit times, and the spectral distinction persists throughout the spacetime’s entire development.

The development proceeds from the composite bundle geometry on $TM \oplus N$ and its curvature Hessian to the explicit cylindrical configuration $\Psi_{\theta,k}$ in Section II and Section III. Section IV establishes the quantitative spectral gap and representation theory, while Section V derives the exact Einstein–Yang–Mills Cauchy development and photon optical geodesics. Section VI details the massive quantum probe response and the optical lattice atom gradiometer detection protocol. Section VII discusses the structural resolution of the Aharonov–Bohm distinction, with full derivations and certified estimates presented in Appendices A and B.

II. CURVATURE ENERGY HESSIAN

Let (M, h) be a connected, smooth, compact, boundaryless Riemannian manifold of dimension d , and let (N, g_N) be a finite-rank real vector bundle over M equipped with a smooth fiber metric g_N . A field configuration is an

ordered pair

$$\Psi = (\nabla^N, K), \quad (7)$$

$$\nabla^N \in \mathcal{A}_{\text{met}}(N, g_N), \quad K \in \Gamma(\text{Sym}^2 T^*M \otimes N),$$

where $\mathcal{A}_{\text{met}}(N, g_N)$ denotes the space of smooth metric connections on (N, g_N) , and K is a smooth symmetric, internal-valued covariant two-tensor. For each vector field $X \in \Gamma(TM)$, contraction yields a bundle homomorphism $K_X: TM \rightarrow N$ defined by $K_X Y = K(X, Y)$. The bundle metrics h and g_N uniquely determine its metric adjoint $K_X^\dagger: N \rightarrow TM$ by

$$h(K_X^\dagger \xi, Y) = g_N(\xi, K_X Y), \quad \forall Y \in TM, \xi \in N. \quad (8)$$

Together with the Levi-Civita connection ∇^h on (TM, h) , the configuration Ψ defines a composite connection $\nabla_\Psi^\oplus$ on the Whitney sum bundle $TM \oplus N$,

$$\nabla_{\Psi, X}^\oplus = \begin{pmatrix} \nabla_X^h & -K_X^\dagger \\ K_X & \nabla_X^N \end{pmatrix}. \quad (9)$$

Because ∇^h and ∇^N are metric connections on (TM, h) and (N, g_N) respectively, and the off-diagonal blocks satisfy

$$h(-K_X^\dagger \xi, Y) + g_N(\xi, K_X Y) = 0, \quad (10)$$

the composite connection $\nabla_\Psi^\oplus$ strictly preserves the direct sum metric $h \oplus g_N$.

A. Geometric Tangent Variations

Let $\mathcal{F}_\Psi^\oplus$ be the curvature two-form of $\nabla_\Psi^\oplus$, taking values in the skew-adjoint endomorphisms $\mathfrak{so}(TM \oplus N)$. In terms of the block decomposition, the curvature splits into three distinct geometric sectors,

$$\mathcal{F}_\Psi^\oplus(X, Y)_{TM, TM} = R^h(X, Y) - K_X^\dagger K_Y + K_Y^\dagger K_X, \quad (11)$$

$$\mathcal{F}_\Psi^\oplus(X, Y)_{N, TM} = (\nabla_X K)_Y - (\nabla_Y K)_X, \quad (12)$$

$$\mathcal{F}_\Psi^\oplus(X, Y)_{N, N} = F^{\nabla^N}(X, Y) - K_X K_Y^\dagger + K_Y K_X^\dagger, \quad (13)$$

corresponding respectively to Gauss-type curvature, Codazzi-type differential constraints, and Ricci-type curvature on the combined bundle.

Using the pointwise Hilbert-Schmidt inner product on two-forms with values in $\mathfrak{so}(TM \oplus N)$, the curvature energy functional is

$$\Phi(\Psi) = \frac{1}{2} \int_M |\mathcal{F}_\Psi^\oplus|^2 dV_h. \quad (14)$$

The configuration space $\mathcal{C}$ carries a natural L^2 Riemannian metric G_Ψ . For variations $v = (a, \kappa) \in \Omega^1(M, \mathfrak{so}(N)) \oplus \Gamma(\text{Sym}^2 T^*M \otimes N)$,

$$G_\Psi(v, v') = \int_M (\langle a, a' \rangle + \langle \kappa, \kappa' \rangle) dV_h. \quad (15)$$

The differential of the curvature energy defines the metric-dual response vector field $\mathcal{V}_\Psi$ by the Riesz representation

$$D\Phi_\Psi(v) = G_\Psi(\mathcal{V}_\Psi, v), \quad \forall v \in T_\Psi \mathcal{C}. \quad (16)$$

Orthogonal projection onto the connection and shape coordinates yields the split response $\mathcal{V}_\Psi = (\mathcal{V}_{\nabla, \Psi}, \mathcal{V}_{K, \Psi})$, where $\mathcal{V}_{\nabla, \Psi}$ acts as the internal gauge response and $\mathcal{V}_{K, \Psi}$ acts as the shape response.

B. Charged versus Singlet Sector Decomposition

Whenever the image subbundle

$$L_K = \text{im } K = \text{span}\{K(X, Y) : X, Y \in TM\} \subset N \quad (17)$$

is a smooth, oriented, metric line subbundle preserved by ∇^N (that is, $\nabla_X^N \Gamma(L_K) \subset \Gamma(L_K)$ for all $X \in TM$), the total bundle decomposes orthogonally as

$$TM \oplus N = (TM \oplus L_K) \oplus E_K, \quad E_K = L_K^\perp \subset N. \quad (18)$$

Variations of K within L_K correspond to the rank-one gravitational sector of the first summand, while variations into E_K represent charged fluctuations.

Let $P_K: N \rightarrow L_K$ and $P_E: N \rightarrow E_K$ be the orthogonal bundle projectors, with $P_K + P_E = \text{Id}_N$. For a tangent variation $v = (a, \kappa)$, the source-induced charged and singlet projectors are defined by

$$\Pi_{\text{ch}, K}(a, \kappa) = (P_K a P_E + P_E a P_K, P_E \kappa), \quad (19)$$

$$\Pi_{0, K}(a, \kappa) = (P_K a P_K + P_E a P_E, P_K \kappa). \quad (20)$$

These projectors are mutually orthogonal in G_Ψ , satisfy $\Pi_{\text{ch}, K} + \Pi_{0, K} = \text{Id}_{T_\Psi \mathcal{C}}$, and are determined entirely by the configuration Ψ and its bundle geometry, without reference to any spectral calculation. When $\nabla^N P_K = 0$, both projectors commute with the gauge action and preserve the gauge-vertical subbundle.

C. Gauge-fixed transverse Hessian

The gauge group $\mathcal{G} = \text{Aut}(N, g_N)$ acts on configurations by

$$u \cdot (\nabla^N, K) = (u \nabla^N u^{-1}, uK). \quad (21)$$

The infinitesimal gauge-vertical space at Ψ is

$$\mathcal{V}_\Psi = \{(-D_{\nabla^N} \xi, \xi K) : \xi \in \Gamma(\mathfrak{so}(N))\}. \quad (22)$$

Let $\mathcal{H}_\Psi$ be the Hilbert space completion of the smooth variations $T_\Psi \mathcal{C}$ under the L^2 metric G_Ψ , and let $\bar{\mathcal{V}}_\Psi$ be the closure of $\mathcal{V}_\Psi$. The physical state space is the orthogonal gauge slice

$$\mathcal{H}_{\text{phys}, \Psi} = \mathcal{H}_\Psi / \bar{\mathcal{V}}_\Psi \cong \bar{\mathcal{V}}_\Psi^\perp. \quad (23)$$

This construction of the quotient space $\mathcal{C}/\mathcal{G}$ and its local orthogonal slice generalizes the classical geometry of gauge-orbit manifolds and slice theorems established by Singer [18], Mitter and Viallet [19], and Atiyah and Bott [20] for Yang–Mills connections, extended here to accommodate the coupled tensorial shape field K .

At a curvature-flat critical configuration Ψ (where $\mathcal{F}_\Psi^\oplus = 0$), the second variation of the curvature energy Φ is given by the quadratic form

$$Q_\Psi(v) = \|\mathcal{L}_\Psi v\|_{L^2}^2, \quad \mathcal{L}_\Psi v = D_{\nabla_\Psi^\oplus} X_v, \quad (24)$$

where $X_v \in \Omega^1(M, \mathfrak{so}(TM \oplus N))$ is the total one-form fluctuation associated with $v = (a, \kappa)$, explicitly

$$(X_v)_X = \begin{pmatrix} 0 & -\kappa_X^\dagger \\ \kappa_X & a_X \end{pmatrix}. \quad (25)$$

By Kato’s first representation theorem [21, Chap. VI, Sec. 2, Theorem 2.1], the densely defined, closed, nonnegative quadratic form Q_Ψ on $\mathcal{H}_{\text{phys}, \Psi}$ uniquely determines a nonnegative self-adjoint operator L_Ψ satisfying

$$Q_\Psi(v, w) = G_\Psi(L_\Psi v, w), \quad (26)$$

for all $v \in \text{dom}(L_\Psi)$ and $w \in \text{dom}(Q_\Psi)$. Because the source projectors $\Pi_{\text{ch}, K}$ and $\Pi_{0, K}$ commute with $\mathcal{L}_\Psi$ and preserve $\mathcal{V}_\Psi^\perp$, the self-adjoint Hessian decomposes as an orthogonal direct sum,

$$L_\Psi = L_\Psi^{\text{ch}} \oplus L_\Psi^0. \quad (27)$$

Thus, the curvature energy functional Φ intrinsically determines both the classical field response and the spectral theory of its gauge-reduced Hessian.

III. COAXIAL CYLINDRICAL SOLENOID MODEL

The field-response theory of Section II specializes to the coaxial cylindrical domain surrounding an enclosed magnetic solenoid,

$$M = [R_1, R_2] \times S^1 \times [0, L], \quad N = M \times \mathbb{R}^3, \quad (28)$$

equipped with the standard flat cylindrical Riemannian metric

$$h = dr^2 + r^2 d\phi^2 + dz^2, \quad (29)$$

and trivial bundle metric $g_N = \sum_{a=1}^3 (\theta^a)^2$. The inner boundary $r = R_1$ represents the superconducting shield enclosing the magnetic core, while the outer boundary $r = R_2$ represents the vacuum chamber wall. The coordinate $\phi \in [0, 2\pi)$ parameterizes the azimuthal circle around the flux core. Let $(e_r, e_\phi, e_z) = (\partial_r, r^{-1}\partial_\phi, \partial_z)$ be the standard orthonormal frame for TM , and let (e_1, e_2, e_3) be the constant orthonormal frame for the internal fiber N .

In the fiber N , the generator of rotations in the $e_2 \wedge e_3$ plane is

$$J = J_{e_2 e_3} \in \mathfrak{so}(3), \quad J e_1 = 0, \quad J e_2 = e_3, \quad J e_3 = -e_2. \quad (30)$$

The two-parameter aligned family of configurations $\Psi_{\theta, k} = (A_\theta, K_k)$ is defined by

$$A_\theta = \theta d\phi J = \frac{\theta}{r} \omega^\phi J, \quad K_k = k (d\phi)^2 e_1 = \frac{k}{r^2} (\omega^\phi)^2 e_1, \\ \theta \in \mathbb{R}/\mathbb{Z}, \quad k \in \mathbb{R} \setminus \{0\}, \quad (31)$$

where $\omega^\phi = r d\phi$ is the orthonormal azimuthal dual form. Both the internal connection A_θ and the spatial profile of K_k are directed along the azimuthal circle coordinate ϕ , aligning the gauge and gravitational degrees of freedom along the closed loop encircling the solenoid.

A. Boundary Data for Flat Connections

Every member of the aligned cylindrical family (31) has identically vanishing bundle curvature $\mathcal{F}_\Psi^\oplus = 0$ in the exterior region M . First, because A_θ is a closed one-form on M ($d(d\phi) = 0$ for $r \geq R_1 > 0$) with values in an abelian subalgebra $\mathbb{R}J \subset \mathfrak{so}(3)$, its Yang–Mills field strength vanishes identically,

$$F_{A_\theta} = dA_\theta + \frac{1}{2}[A_\theta \wedge A_\theta] = 0. \quad (32)$$

Second, because $J e_1 = 0$, the covariant derivative vanishes identically,

$$D_{A_\theta} K_k = dK_k + A_\theta \wedge K_k = 0. \quad (33)$$

Finally, for any vectors $X, Y \in TM$, the linear map $(K_k)_X: TM \rightarrow N$ is a multiple of the single rank-one map $d\phi \otimes e_1$. Consequently, the quadratic commutator terms vanish,

$$-K_X^\dagger K_Y + K_Y^\dagger K_X = 0, \quad -K_X K_Y^\dagger + K_Y K_X^\dagger = 0. \quad (34)$$

Substituting these relations into the Gauss, Codazzi, and Ricci sectors (11)–(13) yields

$$\mathcal{F}^\oplus(A_\theta, K_k) = 0, \quad \Phi(A_\theta, K_k) = 0. \quad (35)$$

Thus, every configuration in the aligned family is an absolute global minimum of the curvature energy Φ , with vanishing energy and vanishing classical response $\mathcal{Y}_{\Psi_{\theta, k}} = 0$.

Now consider any contractible open region $U \subset M$ that does not completely encircle the inner cylinder $r = R_1$. On U , the smooth gauge transformation

$$u_{\theta, U}(r, \phi, z) = \exp(\theta \phi J) \in C^\infty(U, \text{SO}(3)) \quad (36)$$

is single-valued. Under this transformation,

$$u_{\theta, U} A_\theta u_{\theta, U}^{-1} - du_{\theta, U} u_{\theta, U}^{-1} = 0, \quad u_{\theta, U} K_k = K_k, \quad (37)$$

since $\exp(\theta\phi J)e_1 = e_1$. Hence, on every contractible patch,

$$(A_\theta, K_k) \sim (0, K_k). \quad (38)$$

Any local measurement of field strength, curvature invariants, or gauge-covariant local tensors cannot distinguish (A_θ, K_k) from $(0, K_k)$. The parameter θ is purely global.

B. Commuting Gauge-Gravitomagnetic Holonomies

The global topological and geometric content of the configuration resides in the holonomy of $\nabla_{\Psi_{\theta,k}}^\oplus$ around the closed circular cycle $\gamma(s) = (r_0, s, z_0)$ for $s \in [0, 2\pi]$ encircling the inner solenoid core at radius $r_0 \in [R_1, R_2]$.

On the six-dimensional fiber $TM \oplus N$, the composite connection along the azimuthal direction is

$$\nabla_{\Psi,\phi}^\oplus = \theta J_{e_2 e_3} + k J_{e_\phi e_1}. \quad (39)$$

Because $J_{e_2 e_3}$ acts exclusively on the internal charged subspace $E_K = \text{span}\{e_2, e_3\}$ and $J_{e_\phi e_1}$ acts exclusively on the preserved plane $Q_K = \text{span}\{e_\phi, e_1\}$, these two generators commute,

$$[J_{e_2 e_3}, J_{e_\phi e_1}] = 0. \quad (40)$$

Consequently, parallel transport around the closed circular path γ yields a block-diagonal joint holonomy,

$$H^\oplus(\theta, k) = H^g(\theta) \oplus H^K(k) \oplus I_{\text{span}\{e_r, e_z\}}, \quad (41)$$

where

$$H^g(\theta) = \exp(-2\pi\theta J_{e_2 e_3})|_{E_K} \in \text{SO}(2), \quad (42)$$

$$H^K(k) = \exp(2\pi k J_{e_\phi e_1})|_{Q_K} \in \text{SO}(2). \quad (43)$$

This block decomposition explains how different physical probes interact with the same underlying field.

Charged matter probe (electron).—In physical spacetime, an electron traverses the spatial loop (governed by spacetime geodesics) while carrying an internal representation of the gauge group $\text{SO}(3)$. In the idealized test-particle approximation ($G \rightarrow 0$, $m_e \rightarrow 0$), neglecting the metric geodesic phase isolates parallel transport in the internal fiber E_K , yielding the gauge block $H^g(\theta)$ as a geometric holonomy phase [4–6]. In the fundamental representation on E_K , its conjugacy-invariant character is

$$\chi_A(\theta) = \frac{1}{2} \text{Tr}_{E_K} H^g(\theta)^m = \cos(2\pi m\theta). \quad (44)$$

Neutral geometric probe (photon and quantum matter waves).—A photon has zero internal charge ($q = 0$) and couples to the spacetime metric and spatial tangent frames. Its transport along null geodesics around the field detects the spatial metric warping parameterized by the shape

tensor K_k , reading the geometric holonomy $H^K(k)$ with character

$$\chi_K(k) = \frac{1}{2} \text{Tr}_{Q_K} H^K(k)^m = \cos(2\pi mk). \quad (45)$$

More generally, neutral quantum matter waves circulating a closed loop acquire a geometric phase proportional to the closed-loop proper-time difference $\Delta\phi = \frac{mc^2}{\hbar} \oint d\tau = \frac{mc}{\hbar} \oint g_{0i} dx^i$, as originally formulated for matter waves in gravitational fields by Dowker [22] and Stodolsky [23]. In terrestrial laboratories, Earth’s rotating mass distribution generates an ambient, static off-diagonal gravitomagnetic vector potential $\mathbf{A}_{\text{grav}}^\oplus = \frac{2G}{c^3 r^3} (\mathbf{J}_\oplus \times \mathbf{r})$ ($k_\oplus \approx 7 \times 10^{-10}$) whose closed-loop metric holonomy constitutes a gravitational Aharonov–Bohm effect, recently shown to be measurable by macroscopic superfluid ^4He quantum gyrometry with 10^{-35} s proper-time sensitivity [24]. Beyond this ambient laboratory background, the solenoid’s own confined electromagnetic field energy $T_{\mu\nu}^{\text{EM}} \propto B^2$ actively deforms exterior spacetime, sourcing the dynamic gravitomagnetic holonomy $H^K(k)$. Thus, $H^g(\theta)$ and $H^K(k)$ represent exact physical counterparts, where the former is the classic electromagnetic gauge holonomy, while the latter is the field’s own gravitomagnetic and metric holonomy.

Coupled field fluctuations.—In contrast to external probes that sample either H^g or H^K in isolation, the field’s own fluctuation space $\Omega^1(M, E_K) \otimes \Gamma(\text{Sym}^2 T^*M \otimes E_K)$ couples spatial tangent vectors to internal fibers, reflecting the foundational role of loop holonomies and path-ordered factors in unified gauge and gravitational theories [11, 12]. Under the joint holonomy (41), these fluctuations transform in the *tensor-product representation* $H^g \otimes H^K$. In any tensor-product representation, group characters multiply,

$$\begin{aligned} \chi_{\text{mixed}}(H^\oplus(\theta, k)^m) &= \chi_A(\theta) \chi_K(k) \\ &= \cos(2\pi m\theta) \cos(2\pi mk). \end{aligned} \quad (46)$$

Thus, the product of characters is not an artifact of Fourier summation, but the fundamental signature of the tensor-product representation carried by the field’s own coupled degrees of freedom. These parallel transport behaviors are illustrated geometrically in Fig. 1.

C. Dimensional reduction on the cylinder

Because the background configuration $\Psi_{\theta,k}$ in (31) is invariant under translations along the cylinder axis, separation of variables yields standing waves $Z_{n_z}(z) = \sqrt{2/L} \cos(n_z \pi z/L)$ under physical Neumann endcap boundary conditions (or $\sin(n_z \pi z/L)$ under Dirichlet conditions), with discrete axial wavenumbers $k_z = n_z \pi/L$ ($n_z \in \mathbb{N}_0$).

The orthogonal projector onto the zero axial momentum

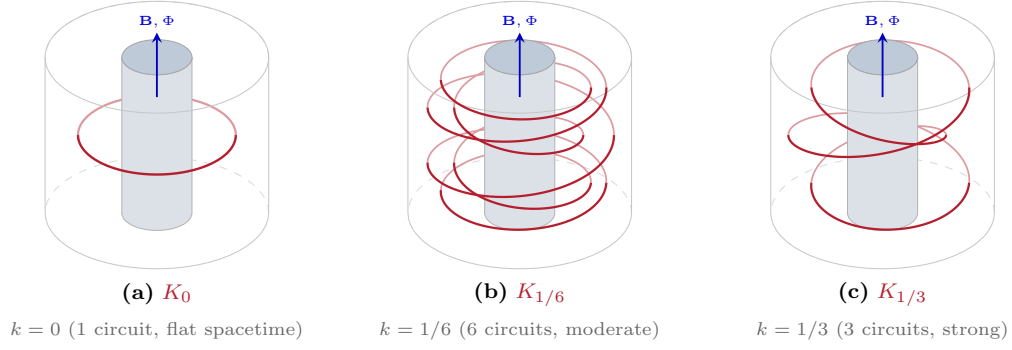


FIG. 1. Parallel transport orbits on the coaxial cylinder encircling the central solenoid core under dynamically increasing gravitomagnetic metric holonomy K_k (bold red curves). Each panel depicts the coaxial vacuum domain $r \in [R_1, R_2]$ surrounding the shielded solenoid core (shaded, carrying axial magnetic flux $\mathbf{B}, \Phi$), with panels **(a)** K_0 ($k = 0$, single closed horizontal ring in flat, unwarped spacetime); **(b)** $K_{1/6}$ ($k = 1/6$, 6 circuits to close 2π , moderate field gravitomagnetism); **(c)** $K_{1/3}$ ($k = 1/3$, 3 circuits, strong field gravitomagnetism with doubled vertical pitch). In contrast to Earth's static ambient laboratory background ($k_\oplus \sim 7 \times 10^{-10}$), the field's gravitomagnetic holonomy is dynamic and sourced by the confined stress-energy $T_{\mu\nu}^{\text{EM}} \propto B^2$, transforming the spatial parallel transport from a planar ring into a multi-circuit helix as field energy increases.

subspace $k_z = 0$ ($n_z = 0$) is defined by

$$P_z v = \frac{1}{L} \int_0^L v(r, \phi, z) dz. \quad (47)$$

The projector P_z commutes with the gauge-reduced Hessian $L_{\theta, k}$ and with the internal projectors $\Pi_{\text{ch}, K}$ and $\Pi_{0, K}$. It isolates the closed subspace of modes depending solely on the radial coordinate $r \in [R_1, R_2]$ and the azimuthal angle $\phi \in [0, 2\pi)$.

IV. HEAT-TRACE SPECTRAL GAP

The spectrum of the gauge-reduced Hessian $L_{\theta, k}$ on the coaxial cylinder M diagonalizes under the complexified J -eigenbasis $q_\sigma = \frac{1}{\sqrt{2}}(e_2 - i\sigma e_3)$ ($\sigma = \pm 1$) of the charged fiber $E_K = \text{span}\{e_2, e_3\}$, satisfying $Jq_\sigma = i\sigma q_\sigma$. On an azimuthal mode $e^{im\phi} q_\sigma$ ($m \in \mathbb{Z}$), the background connection $A_\theta = \theta d\phi J$ acts by covariant derivative $D_{A_\theta} = i(m + \sigma\theta)d\phi$. The gauge-covariant azimuthal momentum is thus

$$p_{m, \sigma} = m + \sigma\theta. \quad (48)$$

A. Bessel mode spectrum in the charged sector

Each charged fluctuation in the σ -eigenspace consists of three connection components a_i and six symmetric shape components κ_{ij} . Separating variables into cylindrical modes $R(r)e^{im\phi} Z_{n_z}(z)$ with axial standing waves $Z_{n_z}(z) = \sqrt{2/L} \cos(n_z \pi z / L)$ ($k_z = n_z \pi / L$), the radial profile $R(r)$ satisfies Bessel's differential equation of shifted order $\nu = |p_{m, \sigma}| = |m + \sigma\theta|$,

$$r^2 \frac{d^2 R}{dr^2} + r \frac{dR}{dr} + (k_r^2 r^2 - (m + \sigma\theta)^2) R(r) = 0. \quad (49)$$

On the annular interval $r \in [R_1, R_2]$, the general solution is $R(r) = C_1 J_\nu(k_r r) + C_2 Y_\nu(k_r r)$. Imposing coaxial boundary conditions at the superconducting solenoid shield $r = R_1$ and outer chamber $r = R_2$ yields the secular equation

$$J_\nu(k_r R_1) Y_\nu(k_r R_2) - J_\nu(k_r R_2) Y_\nu(k_r R_1) = 0. \quad (50)$$

By Watson's monotonicity theorem [25], every radial eigenvalue root $k_{r, n}^{(\nu)}$ is a strictly increasing function of $\nu = |m + \sigma\theta|$, guaranteeing that shifting θ dynamically shifts the radial spectrum.

Appendix A decouples the connection-shape system into block-diagonal form and establishes the characteristic polynomial of the mode stiffness Gram matrix $G(p, k)$,

$$\det(\lambda I - G(p, k)) = -\lambda^2 (2p^2 - \lambda)^3 \times ((p - k)^2 - \lambda)^2 ((p + k)^2 - \lambda)^2. \quad (51)$$

The two zero modes correspond to one gauge-vertical direction along the internal gauge orbit and one non-gauge physical zero mode. After quotienting by the gauge-vertical direction, the nonzero physical eigenvalues and their multiplicities are

$$\begin{array}{c|ccc} \lambda & 2p^2 & (p - k)^2 & (p + k)^2 \\ m(\lambda) & 3 & 2 & 2 \end{array} \quad (52)$$

The eigenvalue families are illustrated in Fig. 2 across three gravitomagnetic configurations matching Fig. 1. The $2p^2$ triplet represents purely transverse charged shape fluctuations; the doublets $(p \mp k)^2$ are equal-weight hybrids of charged connection and azimuthal shape fluctuations coupled by the amplitude k .

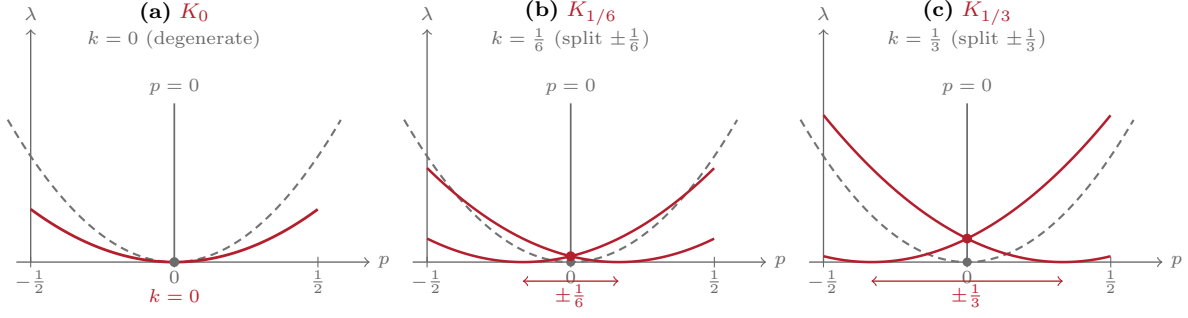


FIG. 2. The charged eigenvalue families of Eq. (52) plotted over a fundamental cell of covariant momentum $p \in [-1/2, 1/2]$ under dynamically increasing gravitomagnetic holonomy $k \in \{0, 1/6, 1/3\}$, matching the three columns of Fig. 1. Dashed curves depict the k -independent transverse triplet $2p^2$, while bold red curves show the coupled gravitomagnetic doublets $(p \mp k)^2$. The vertical gray comb marks the ground-state momentum $p = 0$, with panels (a) K_0 ($k = 0$, degenerate doublets $(p - 0)^2 = p^2$, sampling $\lambda = 0$); (b) $K_{1/6}$ ($k = 1/6$, doublets split by $\pm 1/6$, lifting the sampled doublet eigenvalue to $\lambda = 1/36$); (c) $K_{1/3}$ ($k = 1/3$, doublets split widely by $\pm 1/3$, lifting the sampled doublet eigenvalue to $\lambda = 1/9$). As the field's stress-energy increases, the gravitomagnetic doublets pull apart, dynamically lifting the physical mode stiffness and opening the spectral gap.

B. Poisson Winding Expansion

Because the gauge-reduced Hessian possesses an infinite-dimensional kernel on unregularized cylindrical domains, $\dim \ker L_{\theta,k}^{\text{ch}} = \infty$ (stemming from transverse gauge-invariant modes and non-propagating boundary harmonics), the standard unweighted heat semigroup e^{-tL} is not trace class. In geometric analysis and field theory, regularizing spectral invariants in the presence of zero modes and non-trivial holonomies requires dedicated spectral weighting, as in Ray and Singer's analytic torsion [13], the Cheeger–Müller theorem [14, 15], flat-bundle spectral invariants [16, 17], and modern massive regularizations of topological field theories [26, 27]. Here, the canonical minimal weight function $h_t(x) = xe^{-tx} = -\partial_t e^{-tx}$ ($t > 0$) regularizes the trace by vanishing identically at $x = 0$, strictly isolating the positive spectrum. Applying the axial projector P_z , the weighted heat trace evaluates to the absolutely convergent angular mode sum

$$\begin{aligned} \Xi_{\theta,k}^{\text{ch}}(t) &= \text{Tr} \left(P_z L_{\theta,k}^{\text{ch}} e^{-tL_{\theta,k}^{\text{ch}}} \right) \\ &= \sum_{\sigma=\pm 1} \sum_{m \in \mathbb{Z}} \left[6p_{m,\sigma}^2 e^{-2tp_{m,\sigma}^2} \right. \\ &\quad + 2(p_{m,\sigma} - k)^2 e^{-t(p_{m,\sigma} - k)^2} \\ &\quad \left. + 2(p_{m,\sigma} + k)^2 e^{-t(p_{m,\sigma} + k)^2} \right]. \end{aligned} \quad (53)$$

For any $c > 0$ and $w \in \mathbb{Z}$, let $\mathcal{A}_c(w, t)$ denote the Fourier transform of the weighted Gaussian derived in Appendix B,

$$\mathcal{A}_c(w, t) = \sqrt{\frac{\pi}{c}} t^{-3/2} \left(\frac{1}{2} - \frac{\pi^2 w^2}{ct} \right) \exp \left(-\frac{\pi^2 w^2}{ct} \right). \quad (54)$$

Applying Poisson summation to each term in (53) transforms the discrete angular mode sum into the dual winding

expansion encircling the cylindrical solenoid,

$$\begin{aligned} \Xi_{\theta,k}^{\text{ch}}(t) &= \sum_{w \in \mathbb{Z}} \left[6\mathcal{A}_2(w, t) \cos(2\pi w\theta) \right. \\ &\quad \left. + 8\mathcal{A}_1(w, t) \cos(2\pi w\theta) \cos(2\pi wk) \right]. \end{aligned} \quad (55)$$

The $w = 0$ term is the zero-winding contribution, which is entirely independent of both holonomies,

$$\Xi^{(0)}(t) = \frac{\sqrt{\pi}}{2t^{3/2}} \left(3\sqrt{2} + 8 \right). \quad (56)$$

For each nonzero winding $w \neq 0$ encircling the solenoid, the trace contains the joint term

$$W_w^{\theta k}(t) = 8\mathcal{A}_1(w, t) \cos(2\pi w\theta) \cos(2\pi wk), \quad (57)$$

which is precisely proportional to the product of the gauge character $\chi_A(\theta)$ and the geometric character $\chi_K(k)$.

C. Strict positivity of the spectral gap

The behavior of the total trace follows from the complete w -th winding coefficient,

$$B_w(k; t) = 6\mathcal{A}_2(w, t) + 8\mathcal{A}_1(w, t) \cos(2\pi wk). \quad (58)$$

Appendix B establishes that at $t = 1$, for every $w \neq 0$,

$$\frac{|\mathcal{A}_1(w, 1)|}{|\mathcal{A}_2(w, 1)|} < \frac{3}{4}. \quad (59)$$

Since $\mathcal{A}_2(w, 1) < 0$ for all $w \neq 0$, it follows that $B_w(k; 1) < 0$ strictly for all $k \in \mathbb{R}$. Consequently, evaluating the difference between the unshifted configuration $\theta = 0$ and the half-integer flux $\theta = 1/2$ yields

$$\Xi_{0,k}^{\text{ch}}(1) - \Xi_{1/2,k}^{\text{ch}}(1) = 2 \sum_{w \in 2\mathbb{Z}+1} B_w(k; 1) < 0. \quad (60)$$

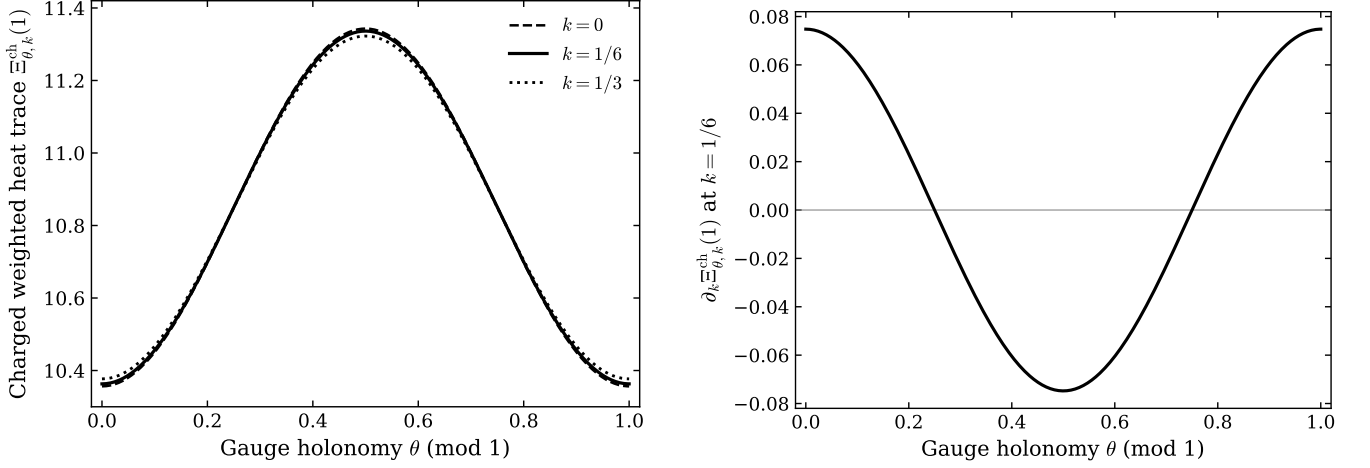


FIG. 3. Left: The charged weighted heat trace $\Xi_{\theta,k}^{\text{ch}}(1)$ as a function of the gauge holonomy parameter $\theta \in [0, 1]$ for shape amplitudes $k = 0, 1/6, 1/3$. The trace exhibits a strictly positive spectral gap between the unshifted state $\theta = 0$ and the half-integer flux state $\theta = 1/2$. Right: The gravitational derivative $\partial_k \Xi_{\theta,k}^{\text{ch}}(1)$ along the preserved rank-one shape deformation.

Hence, $\Xi_{1/2,k}^{\text{ch}}(1) > \Xi_{0,k}^{\text{ch}}(1)$ holds strictly for every non-zero shape amplitude k .

Evaluating the exact sums in Appendix B using Gaussian tail bounds at $k = 1/6$ and $t = 1$ yields

$$\Xi_{0,1/6}^{\text{ch}}(1) = 10.3631980884 \dots, \quad (61)$$

$$\Xi_{1/2,1/6}^{\text{ch}}(1) = 11.3363159945 \dots, \quad (62)$$

which establishes the quantitative spectral gap,

$$\Xi_{1/2,1/6}^{\text{ch}}(1) - \Xi_{0,1/6}^{\text{ch}}(1) > 0.9731179060. \quad (63)$$

D. Cancellation of the neutral singlet sector

Analysis of the neutral singlet sector $L_{\theta,k}^0$ in Appendix A confirms that this separation is an authentic property of the complete physical Hessian $L_{\theta,k}$ rather than an artifact of charged projection. The nonzero singlet spectrum on angular mode m is

$$\text{Spec}_+(L_{m,k}^0) = \{m^2 \text{ (mult 4)}, 2(m^2 + k^2) \text{ (mult 3)}\}. \quad (64)$$

Remarkably, the singlet eigenvalues depend solely on m and k , with zero dependence on the gauge holonomy θ . The singlet contribution to the weighted heat trace is

$$N_k(t) = \sum_{m \in \mathbb{Z}} \left[4m^2 e^{-tm^2} + 6(m^2 + k^2) e^{-2t(m^2 + k^2)} \right]. \quad (65)$$

By the direct sum decomposition (27), the total physical trace is

$$\Xi_{\theta,k}(t) = \Xi_{\theta,k}^{\text{ch}}(t) + N_k(t). \quad (66)$$

Because $N_k(t)$ is strictly independent of θ , the difference between any two gauge holonomies at fixed k is identical

for the charged trace and the complete trace,

$$\Xi_{1/2,k}(t) - \Xi_{0,k}(t) = \Xi_{1/2,k}^{\text{ch}}(t) - \Xi_{0,k}^{\text{ch}}(t). \quad (67)$$

The distinction (63) survives completion to the full physical Hessian without modification.

E. Gravitational Response Stability

Differentiating the absolutely convergent winding sum (55) with respect to the shape amplitude k yields the gravitational spectral response,

$$\partial_k \Xi_{\theta,k}^{\text{ch}}(t) = -16\pi \sum_{w \in \mathbb{Z}} w \mathcal{A}_1(w, t) \cos(2\pi w \theta) \sin(2\pi w k). \quad (68)$$

At $\theta = k = 1/6$ and $t = 1$, the first Fourier coefficient in θ is nonzero, yielding the quantitative value

$$\partial_k \Xi_{1/6,1/6}^{\text{ch}}(1) = 0.0373922800861 \dots > 0. \quad (69)$$

Because the spectrum depends analytically on k and θ , the non-vanishing mixed derivative $\partial_\theta \partial_k \Xi_{\theta,k}^{\text{ch}}(t) \neq 0$ confirms that the field's curvature energy intrinsically couples gauge and spacetime holonomies.

V. EINSTEIN–YANG–MILLS SPACETIME EVOLUTION

The aligned configurations (A_θ, K_k) are not merely static critical points of an energy functional; they arise as exact initial data for the coupled Einstein–Yang–Mills system in four spacetime dimensions.

A. Cauchy Constraint Formulation

Consider the three-dimensional spatial Cauchy slice $\Sigma = [R_1, R_2] \times S^1 \times [0, L]$ surrounding the solenoid, equipped with the flat cylindrical metric $h_0 = dr^2 + r^2 d\phi^2 + dz^2$. The unit section $e_1 \in \Gamma(L_K)$ identifies the shape tensor K_k with a spatial extrinsic curvature tensor directed along the azimuthal circle,

$$\kappa_0 = \langle K_k, e_1 \rangle = k (d\phi)^2 = \frac{k}{r^2} (rd\phi)^2, \quad (70)$$

which is symmetric and has trace $\text{tr}_{h_0} \kappa_0 = k/r^2$. The initial gauge data consists of the flat connection $A_\theta = \theta d\phi$ and vanishing initial electric field $E_0 = 0$.

Direct substitution confirms that the Cauchy data $(h_0, \kappa_0, A_\theta, E_0)$ strictly satisfies the vacuum Einstein–Yang–Mills constraint equations [28].

Yang–Mills Gauss constraint.—Because $E_0 = 0$ and the matter currents vanish,

$$D_{A_\theta} * E_0 = 0. \quad (71)$$

Gravitational momentum constraint.—In the orthonormal frame (e_r, e_ϕ, e_z) , κ_0 has components $\kappa_{0,\hat{\phi}\hat{\phi}} = k/r^2$. The divergence evaluates to $\nabla^j (\kappa_{0,ij} - (\text{tr}_{h_0} \kappa_0) h_{0,ij}) = 0 = 8\pi G T_{0i}$, satisfying the momentum constraint everywhere in the vacuum exterior $r \geq R_1$.

Gravitational Hamiltonian constraint.—For flat cylindrical space $R_{h_0} = 0$. Because κ_0 is rank-one along ϕ , its squared norm is $|\kappa_0|_{h_0}^2 = (k/r^2)^2$, which exactly equals $(\text{tr}_{h_0} \kappa_0)^2 = (k/r^2)^2$. Furthermore, since $F_{A_\theta} = 0$ and $E_0 = 0$, the Yang–Mills energy-momentum tensor vanishes identically in the exterior, $T_{00} = 0$. Thus,

$$R_{h_0} - |\kappa_0|_{h_0}^2 + (\text{tr}_{h_0} \kappa_0)^2 = 0 - \frac{k^2}{r^4} + \frac{k^2}{r^4} = 0 = 16\pi G T_{00}. \quad (72)$$

The initial data satisfies all constraints without approximations or source terms.

B. Lorentzian spacetime evolution

Let τ denote the synchronous proper time coordinate. Define the azimuthal scale factor

$$a(\tau) = 1 - k\tau, \quad I_k = \{\tau \in \mathbb{R} : a(\tau) > 0\}. \quad (73)$$

On the four-dimensional manifold $M_k = I_k \times \Sigma$, the exact Lorentzian spacetime metric and gauge potential are

$$\begin{aligned} \mathbf{g}_k &= -d\tau^2 + dr^2 + a(\tau)^2 r^2 d\phi^2 + dz^2, \\ A_\theta &= \theta d\phi J. \end{aligned} \quad (74)$$

The Yang–Mills field strength vanishes identically on M_k , so $\mathcal{F}_{A_\theta} = 0$ and the Yang–Mills stress-energy tensor is zero throughout the exterior spacetime.

For the spacetime metric $\mathbf{g}_k$, the non-vanishing Christoffel symbols involving τ are $\Gamma_{0\phi}^\phi = \dot{a}/a = -k/a$ and

$\Gamma_{\phi\phi}^0 = a\dot{a}r^2 = -kar^2$. The Riemann curvature tensor components vanish identically,

$$R_{\tau\phi\tau\phi} = -r^2 a\ddot{a} = 0, \quad (75)$$

$$R_{\tau\tau} = -\frac{\ddot{a}}{a} = 0, \quad R_{\phi\phi} = r^2 a\ddot{a} = 0, \quad (76)$$

since $\ddot{a}(\tau) = 0$. Thus, $(M_k, \mathbf{g}_k)$ is a flat Lorentzian spacetime.

On each constant- τ Cauchy hypersurface $\Sigma_\tau = \{\tau\} \times \Sigma$, the induced spatial metric is $h_\tau = dr^2 + a(\tau)^2 r^2 d\phi^2 + dz^2$, and the extrinsic curvature is

$$\kappa_\tau = -\frac{1}{2} \mathcal{L}_{\partial_\tau} h_\tau = ka(\tau)r^2 (d\phi)^2. \quad (77)$$

As $\tau \rightarrow 1/k$, $a(\tau) \rightarrow 0$, marking a physical singularity where the azimuthal circle degenerates. By Choquet-Bruhat and Geroch [29], $(M_k, \mathbf{g}_k)$ is the maximal globally hyperbolic development of the initial data in the smooth coaxial class.

C. Optical Phase Shifts via Null Geodesics

The geometric parameter k is not an ad hoc kinematic artifact; it arises from the stress-energy tensor of the enclosed gauge field through Einstein’s field equations.

Consider an enclosed magnetic flux tube of radius R containing field $\mathbf{B} = B\hat{\mathbf{z}}$, carrying total flux $\Phi = \pi R^2 B$, with vanishing exterior field strength $\mathbf{B} = 0$ for $r > R$. In relativistic electrodynamics, the electromagnetic stress-energy tensor is

$$T^{\mu\nu} = \frac{1}{\mu_0} \left(F^{\mu\alpha} F^\nu{}_\alpha - \frac{1}{4} \eta^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right). \quad (78)$$

Inside the flux core ($r \leq R$), the energy density is $u = T_{00} = B^2/(2\mu_0)$ with transverse magnetic pressure $T^r_r = T^\phi_\phi = B^2/(2\mu_0)$ and longitudinal tension $T^z_z = -B^2/(2\mu_0)$. Integrating over the cross-section yields the linear mass-energy density,

$$\mu = \frac{1}{c^2} \int_0^{2\pi} d\phi \int_0^R r dr T_{00} = \frac{\pi R^2 B^2}{2\mu_0 c^2} = \frac{\Phi^2}{2\pi\mu_0 c^2 R^2}. \quad (79)$$

By Einstein’s field equations $G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}$, this confined mass-energy generates an exterior metric ($r \geq R$) that is locally flat ($R_{\mu\nu} = 0$) but possesses a global conical deficit angle [22, 30, 31] $\Delta\phi_{\text{deficit}} = 8\pi G\mu/c^2$. The dimensionless deficit parameter is

$$\delta = \frac{\Delta\phi_{\text{deficit}}}{2\pi} = \frac{4G\mu}{c^2} = \frac{2GB^2 R^2}{\mu_0 c^4} = \frac{2G\Phi^2}{\pi^2 \mu_0 c^4 R^2}. \quad (80)$$

In cylindrical coordinates (t, r, ϕ, z) , the exterior line element for $r \geq R$ is

$$ds^2 = -c^2 dt^2 + dr^2 + (1 - \delta)^2 r^2 d\phi^2 + dz^2. \quad (81)$$

For a closed cycle γ_1 encircling the flux at radius $r_0 \geq R$, the physical optical circumference is $L_{\text{opt}} = 2\pi(1 - \delta)r_0$, representing a contraction $\Delta L_{\text{opt}} = -2\pi r_0 \delta$. In the dynamical cylindrical representation $(M_k, \mathbf{g}_k)$ with scale factor $a(\tau) = 1 - k\tau$, the extrinsic curvature rate k matches the fractional spatial contraction per characteristic transit time $\tau_{\text{transit}} = 2\pi r_0/c$,

$$k = \frac{c\delta}{2\pi r_0} = \frac{G\Phi^2}{\pi^3 \mu_0 c^3 R^2 r_0}. \quad (82)$$

A neutral optical probe (photon, $q = 0$) follows null geodesics $ds^2 = 0$, accumulating optical transit and phase shifts along spacetime trajectories [23]. For a photon circulating in the azimuthal plane at radius $r_0 \geq R$,

$$-c^2 dt^2 + (1 - \delta)^2 r_0^2 d\phi^2 = 0 \implies \left| \frac{d\phi}{dt} \right| = \frac{c}{(1 - \delta)r_0}. \quad (83)$$

The coordinate transit time for one complete revolution ($\Delta\phi = 2\pi$) is $t_{\text{rev}} = 2\pi(1 - \delta)r_0/c$. Compared to the unperturbed Minkowski transit time $t_0 = 2\pi r_0/c$, the enclosed field induces an exact transit time advance,

$$\Delta\tau = t_{\text{rev}} - t_0 = -\frac{2\pi r_0 \delta}{c} = -\frac{4G\Phi^2 r_0}{\pi \mu_0 c^5 R^2}. \quad (84)$$

For a trajectory grazing the flux boundary ($r_0 = R$), the optical path contraction and transit shift reduce to

$$\Delta L_{\text{opt}} = -\frac{4G\Phi^2}{\pi \mu_0 c^4 R} = -\frac{4\pi G B^2 R^3}{\mu_0 c^4}, \quad (85)$$

$$\Delta\tau = -\frac{4G\Phi^2}{\pi \mu_0 c^5 R}. \quad (86)$$

For an optical probe of vacuum wavelength λ , substituting (86) into $\Delta\phi_{\text{photon}} = (2\pi c/\lambda)|\Delta\tau|$ produces the exact single-pass optical phase shift,

$$\Delta\phi_{\text{photon}} = \frac{8G\Phi^2}{\mu_0 c^4 \lambda R} = \frac{8\pi^2 G B^2 R^3}{\mu_0 c^4 \lambda}. \quad (87)$$

Neutral photons therefore detect the enclosed field through the metric warping and conical deficit induced by its stress-energy tensor.

D. Dynamical stability of the spectral gap

On each Cauchy slice Σ_τ , the orthonormal frame adapted to the metric $h_\tau = dr^2 + a(\tau)^2 r^2 d\phi^2 + dz^2$ is $e_\phi(\tau) = (a(\tau)r)^{-1}\partial_\phi$, $e_r = \partial_r$, $e_z = \partial_z$. In this comoving frame, the gauge-covariant azimuthal momentum scales as $p_{m,\sigma}(\tau) = a(\tau)^{-1}(m + \sigma\theta)$ and the extrinsic curvature amplitude scales as $k(\tau) = a(\tau)^{-1}k$.

Parallel transport and volume rescaling induce a unitary transformation U_τ such that for each sector $\bullet \in \{\text{ch}, 0\}$, the gauge-reduced Hessian at time τ satisfies the operator scaling law,

$$L_{\theta,k}^\bullet(\tau) = a(\tau)^{-2} U_\tau^{-1} L_{\theta,k}^\bullet(0) U_\tau. \quad (88)$$

Applying the scaling identity to the weighted heat trace yields

$$\Xi_{\theta,k}^{\text{ch}}(t; \tau) = a(\tau)^{-2} \Xi_{\theta,k}^{\text{ch}}\left(\frac{t}{a(\tau)^2}; 0\right). \quad (89)$$

In terms of the comoving spectral scale $t(\tau) = a(\tau)^2 t_0$, the exact invariance follows,

$$a(\tau)^2 \Xi_{\theta,k}^{\text{ch}}(a(\tau)^2 t_0; \tau) = \Xi_{\theta,k}^{\text{ch}}(t_0; 0). \quad (90)$$

Consequently, the normalized comoving spectral distinction between $\theta = 0$ and $\theta = 1/2$ persists without degradation throughout the entire existence interval I_k .

Furthermore, at a fixed laboratory spectral scale $t = 1$, the distinction remains strictly positive over an open time window around $\tau = 0$. Defining $D(s) = \Xi_{1/2,1/6}^{\text{ch}}(s; 0) - \Xi_{0,1/6}^{\text{ch}}(s; 0)$, termwise differentiation of the Fourier sum for $s \in [1/2, 3/2]$ gives $|D'(s)| < 307$. For $k = 1/6$ and $|\tau| \leq 10^{-3}$, the perturbation $|a(\tau)^{-2} - 1| < 10^{-3}$, which guarantees

$$D(a(\tau)^{-2}) > 0.9731179060 - 0.307 > 0.666. \quad (91)$$

By (89), the fixed- $t = 1$ trace difference remains strictly bounded away from zero. The two solutions share identical local curvature, identical zero Yang–Mills stress, and the same spacetime metric $\mathbf{g}_k$, yet their intrinsic field responses remain cleanly separated throughout their physical developments.

VI. MATTER-WAVE INTERFEROMETRIC DETECTION

Physical quantum probes interact with the warped exterior geometry of the enclosed field, enabling an unambiguous tabletop protocol to detect this gravitational field curvature in the laboratory.

A. Comparative Sensitivity of Probe Species

The physical mechanism developed in Section V yields exact formulas for both electromagnetic and geometric phase shifts in terms of laboratory parameters,

$$\Delta\phi_{\text{photon}} = \frac{8G\Phi^2}{\mu_0 c^4 \lambda R} = \frac{8\pi^2 G B^2 R^3}{\mu_0 c^4 \lambda}, \quad (92)$$

$$\Delta\phi_e = \frac{e}{\hbar} \Phi = \frac{\pi e B R^2}{\hbar}. \quad (93)$$

These expressions illuminate a fundamental physical distinction.

Quadratic vs. linear flux scaling.—In a charged matter wave (such as an electron beam), the total phase accumulates both gauge and metric terms, $\Delta\phi_e = \Delta\phi_e^{\text{gauge}} + \Delta\phi_e^{\text{grav}}$. The gauge component $\Delta\phi_e^{\text{gauge}}$ in Eq. (93) is

strictly *linear* in the flux ($\Delta\phi_e \propto \Phi$), reflecting minimal coupling $q \oint A_\mu dx^\mu$. In contrast, the gravitational geodesic phase $\Delta\phi_{\text{photon}}$ (and the metric phase of any matter probe) is strictly *quadratic* in the field ($\propto \Phi^2 \propto B^2$), reflecting the universal coupling of spacetime geometry to the stress-energy tensor $T_{\mu\nu} \propto B^2$. Because the electron's linear gauge phase ($\sim 10^{14}$ rad in Table I) exceeds its own quadratic metric phase ($\sim 10^{-26}$ rad) by forty orders of magnitude, the metric back-reaction is completely masked in electron interferometry. This establishes why neutral probes are physically essential; by setting the linear gauge phase identically to zero, neutral probes eliminate the 10^{40} masking, allowing the pure quadratic gravitational curvature of the field to be observed.

Cross-probe dimensionless ratio.—The ratio of the neutral optical phase to the charged electron phase isolates the gravitational-to-electromagnetic coupling hierarchy:

$$\frac{\Delta\phi_{\text{photon}}}{\Delta\phi_e} = \frac{8\pi G\hbar}{\mu_0 c^4 e \lambda} (BR) = \left(\frac{8G\hbar}{\mu_0 c^4 e} \right) \frac{\Phi}{\lambda R}. \quad (94)$$

Table I reports the evaluated predictions across three physical regimes. While the single-pass photon phase shift $\sim 10^{-32}$ rad is suppressed by the fundamental ratio $G/(\mu_0 c^4) \approx 6.575 \times 10^{-39} \text{ T}^{-2} \text{ m}^{-1}$, cold-atom matter waves in an optical lattice amplify this geometric signal by over twenty-one orders of magnitude, elevating the metric phase into the measurable sub-microradian regime.

B. Cold-Atom Sensitivity Scaling

While photons are massless and traverse the warped metric at the speed of light c , quantum mechanics introduces an enormous enhancement when the probe is replaced by a *massive neutral particle*, such as a neutron, a laser-cooled atom, or a macromolecule. Because the particle carries zero electric charge, it remains completely immune to the 10^{14} rad electromagnetic gauge masking that overwhelms charged matter, while its rest mass couples directly to the warped spacetime metric.

A massive neutral particle couples to spacetime geometry via its relativistic classical action,

$$S[\gamma] = -mc \int_\gamma ds = -mc^2 \int_0^T dt \sqrt{1 - \frac{v^2}{c^2}}. \quad (95)$$

For an orbital trajectory of radius $r_0 \geq R$ traversed at non-relativistic velocity $v \ll c$, the action perturbation induced by the conical deficit δ is $\Delta S \approx (mc^2/v)|\Delta L_{\text{opt}}|$, where $|\Delta L_{\text{opt}}| = 2\pi r_0 \delta$. The accumulated quantum phase shift is

$$\Delta\phi_{\text{massive}} = \frac{\Delta S}{\hbar} = \frac{mc^2}{\hbar v} |\Delta L_{\text{opt}}| = \frac{4\pi G m B^2 R^3}{\mu_0 c^2 \hbar v}. \quad (96)$$

Dividing (96) by the photon phase shift $\Delta\phi_{\text{photon}} = (E_\gamma/\hbar c)|\Delta L_{\text{opt}}|$ yields the *Dual-Factor Amplification The-*

orem,

$$\frac{\Delta\phi_{\text{massive}}}{\Delta\phi_{\text{photon}}} = \underbrace{\left(\frac{mc^2}{E_\gamma} \right)}_{\text{Compton energy boost}} \times \underbrace{\left(\frac{c}{v} \right)}_{\text{Kinetic slowness boost}}. \quad (97)$$

Both factors provide substantial numerical gains. First, the *rest-mass Compton boost* mc^2/E_γ provides an enormous energy advantage: a ^{87}Rb atom has rest-mass energy $m_{\text{Rb}}c^2 \approx 81.0 \text{ GeV}$, exceeding a 1064 nm optical photon (1.165 eV) by a factor of 6.95×10^{10} . Second, the *kinetic slowness boost* c/v provides an exposure-time advantage: in cold-atom fountains or optical lattices, atoms move at non-relativistic velocities $v \sim 1 \text{ cm/s}$ or remain trapped for seconds, spending $c/v \approx 3 \times 10^{10}$ times longer traversing the warped metric than a relativistic photon. Together, these produce an aggregate amplification factor,

$$\frac{\Delta\phi_{\text{atom}}}{\Delta\phi_{\text{photon}}} \approx (6.95 \times 10^{10}) \times (3 \times 10^{10}) \approx 2.09 \times 10^{21}. \quad (98)$$

For a 10 T, 5 cm solenoid, this elevates the geometric phase from 6.1×10^{-33} rad to

$$\Delta\phi_{\text{atom}} \approx 1.28 \times 10^{-11} \text{ rad}. \quad (99)$$

For a heavy macromolecule (10^6 amu) in matter-wave interferometry, the phase shift reaches 1.46×10^{-10} rad.

C. Core Mass versus Field Energy Budget

In any laboratory apparatus, the stress-energy tensor decomposes into the electromagnetic field energy and the structural rest mass of the magnet,

$$T_{\mu\nu} = T_{\mu\nu}^{\text{EM}} + T_{\mu\nu}^{\text{matter}}. \quad (100)$$

For a 10 T field across a 1.57 L core, the confined electromagnetic energy represents a dynamic mass $\Delta M_{\text{field}} = c^{-2} \int u dV \approx 0.70 \mu\text{g}$, whereas the physical coils, structural core, and cryogenic casing have a static rest mass $M_{\text{matter}} \sim 1\text{--}10 \text{ kg}$.

In a light-pulse atom interferometer, the gravitational phase shift between two spatial paths γ_1, γ_2 in a Newtonian potential Φ_{grav} is

$$\Delta\Phi_{\text{grav}} = \frac{m_{\text{atom}}}{\hbar} \int_0^T [\Phi_{\text{grav}}(\gamma_1(t)) - \Phi_{\text{grav}}(\gamma_2(t))] dt. \quad (101)$$

For a source mass $M_{\text{matter}} \approx 1.25 \text{ kg}$ at a separation $d \approx 5 \text{ cm}$, this produces a phase shift

$$\Delta\Phi_{\text{matter}} \sim 0.1 \text{ to } 1.0 \text{ rad}. \quad (102)$$

This macroscopic gravitational phase was directly observed by Overstreet *et al.* [32] in a 10-meter atomic fountain with cold ^{87}Rb atoms and a tungsten cylinder, demonstrating the gravitational Aharonov–Bohm effect

TABLE I. Exact numerical predictions for geometric spacetime warping, optical photon phase shifts, and cold-atom matter-wave signals across representative laboratory magnetic configurations, evaluated with CODATA fundamental constants ($G = 6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$, $c = 2.99792 \times 10^8 \text{ m/s}$, $\hbar = 1.05457 \times 10^{-34} \text{ J s}$, $e = 1.60218 \times 10^{-19} \text{ C}$, $\mu_0 = 4\pi \times 10^{-7} \text{ T m/A}$).

Parameter	Symbol	Unit	Superconducting Solenoid	High-Field Solenoid	Pulsed High-Field Coil
Magnetic field	B	T	10.0	40.0	100.0
Core radius	R	m	0.050 (5 cm)	0.050 (5 cm)	0.010 (1 cm)
Enclosed flux	Φ	Wb	7.854×10^{-2}	3.142×10^{-1}	3.142×10^{-2}
Linear mass density	μ	kg/m	3.477×10^{-12}	5.563×10^{-11}	1.391×10^{-11}
Conical deficit parameter	δ	—	3.288×10^{-39}	5.260×10^{-38}	1.315×10^{-38}
Metric transit time shift	$ \Delta\tau $	s	3.445×10^{-48}	5.512×10^{-47}	2.756×10^{-48}
Photon phase shift (1064 nm)	$\Delta\phi_{\text{photon}}$	rad	6.099×10^{-33}	9.759×10^{-32}	9.759×10^{-33}
Electron AB phase	$\Delta\phi_e$	rad	1.193×10^{14}	4.773×10^{14}	4.771×10^{13}
Cross-probe ratio	$\Delta\phi_{\text{photon}}/\Delta\phi_e$	—	5.111×10^{-47}	2.045×10^{-46}	2.046×10^{-46}
Cold atom phase (^{88}Sr , $v = 1 \text{ cm/s}$)	$\Delta\phi_{\text{atom}}$	rad	1.285×10^{-11}	2.056×10^{-10}	1.028×10^{-11}
Lattice gradiometer signal (10 s, 2ω)	$\delta\phi_{\text{EM}}$	μrad	0.018	0.290	—

with sub-milliradian precision. The experimental foundation for matter-wave interferometry in gravitational fields was established by Colella, Overhauser, and Werner [33] using thermal neutrons, and subsequently pioneered in laser-cooled atoms by Kasevich and Chu [34]. Subsequent precision measurements by Rosi *et al.* [35] and Westphal *et al.* [36] demonstrated that atom interferometers and miniature balances can resolve gravitational couplings from well-characterized laboratory test masses down to millimeter scales. Furthermore, Asenbaum *et al.* [37, 38] directly resolved gravitational tidal curvature phase shifts across macroscopic wavepacket separations and tested the equivalence principle at the 10^{-12} level, establishing cold-atom gradiometry as an established metrological tool [39].

Beyond static Newtonian potentials, matter waves also sense off-diagonal metric connections and gravitomagnetic holonomies. In terrestrial laboratories, Earth's rotation generates a stationary gravitomagnetic vector potential $\mathbf{A}_{\text{grav}} = \frac{2G}{c^3 r^3} (\mathbf{J}_{\oplus} \times \mathbf{r})$. Recently, Ellers *et al.* [24] demonstrated that a macroscopic superfluid ^4He Josephson gyrometer achieves proper-time difference sensitivities reaching 10^{-35} s , resolving frame-dragging through the quantum phase $\Delta\phi = (mc/\hbar) \oint \mathbf{A}_{\text{grav}} \cdot d\mathbf{l}$. In the geometric framework developed here, Earth's gravitomagnetic field provides a natural laboratory realization of the background metric holonomy $H^K(k)$, providing an empirical anchor for the preserved-plane connection in $\nabla_{\Psi}^{\oplus} = A_{\theta} + K_k$.

Together, these experimental milestones demonstrate that quantum matter waves are sensitive probes of space-time geometry, establishing the physical foundation in which gauge fields warp spacetime, photon null geodesics detect optical metric contraction, and quantum matter waves measure metric potential holonomies.

D. Harmonic Lock-In Detection Protocol

To observe the gravitational spacetime curvature induced by the confined electromagnetic field, an experiment must overcome the fact that the static apparatus rest mass exceeds the dynamic field mass by nine orders of magnitude.

To achieve this, the detector device is configured as a *dual-arm optical lattice atom gradiometer* (Fig. 4). Two ensembles of laser-cooled ^{88}Sr atoms in their electronic ground state 1S_0 are trapped in vertical optical lattices operating at the magic wavelength of 813.4 nm. The interacting arm is positioned at distance $d_1 = 6 \text{ cm}$ from the solenoid axis, while the reference arm is placed at $d_2 = 18 \text{ cm}$, establishing a differential baseline of $\Delta d = 12 \text{ cm}$. Because both atomic clouds are interrogated simultaneously by the same optical lattice beams, common-mode environmental perturbations—including laboratory seismic vibrations, building tilt, laser phase noise, and Earth's uniform gravitational acceleration—cancel identically in the differential quantum phase observable,

$$\begin{aligned} \delta\phi_{\text{EM}} &= \phi(d_1) - \phi(d_2) \\ &= \frac{m_{\text{Sr}} c^2}{2\hbar} \int_0^{T_{\text{hold}}} [h_{00}(d_1, t) - h_{00}(d_2, t)] dt. \end{aligned} \quad (103)$$

The detection protocol isolates this metric perturbation from the static apparatus mass through four operational stages.

1. *AC Field Modulation and Frequency Doubling.* The solenoid drive current is modulated harmonically at frequency ω , producing an alternating core flux $B(t) = B_0 \cos(\omega t)$. Because the electromagnetic stress-energy tensor is quadratic in the field, it modulates strictly at the second harmonic 2ω ,

$$T_{\mu\nu}^{\text{EM}}(t, \mathbf{x}) = \frac{1}{2} T_{\mu\nu}^{\text{peak}}(\mathbf{x}) [1 + \cos(2\omega t)]. \quad (104)$$

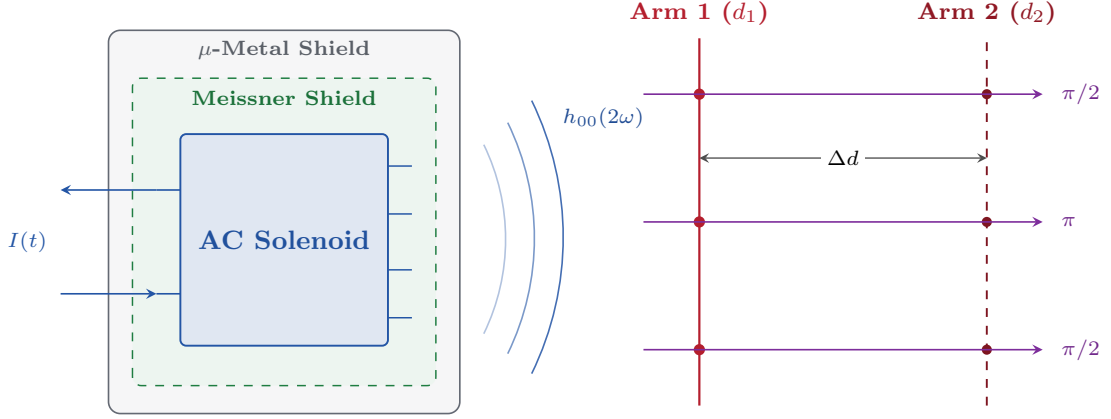


FIG. 4. Proposed tabletop apparatus and dual-arm atom interferometer geometry for measuring the gravitational spacetime curvature of confined electromagnetic fields. An AC-driven solenoid ($B(t) = B_0 \cos \omega t$, with $B_0 = 10\text{--}40\text{ T}$) is enclosed within a cryogenic superconducting Meissner shield and a four-layer μ -metal shield ($S_{\text{total}} > 10^9$), suppressing residual external leakage below 1 nT . Cold ^{88}Sr probe atoms in the spin-0 ground state 1S_0 eliminate first-order Zeeman shifts and magnetic dipole forces, while residual quadratic diamagnetism is suppressed below 10^{-13} rad . A $\pi/2\text{--}\pi\text{--}\pi/2$ pulse sequence interrogates atoms held in optical lattices across a baseline $\Delta d = d_2 - d_1$, sensing the dynamic metric perturbation $h_{00}(2\omega)$.

By symmetry, all odd harmonics vanish identically, ensuring zero dynamic signal at the drive frequency ω .

2. *Spectral Separation of Backgrounds.* The static apparatus rest mass M_{matter} generates an unmodulated DC gravitational background. In contrast, the dynamic metric perturbation oscillating at 2ω ,

$$h_{00}^{(2\omega)}(\mathbf{x}) = \frac{GB_0^2 V}{2\mu_0 c^4 r}, \quad (105)$$

is extracted by phase-locked demodulation at 2ω , rejecting the static Newtonian background by over 100 dB.

3. *Symmetry Suppression of Systematic Magnetic Forces.* To eliminate non-gravitational phase shifts, the probe ensembles employ laser-cooled ^{88}Sr or ^{174}Yb atoms in their spherically symmetric 1S_0 ground state. With zero total electronic and nuclear angular momentum, the atoms have no magnetic dipole moment, eliminating first-order Zeeman shifts and magnetic gradient forces. Residual quadratic diamagnetism is suppressed below 10^{-13} rad by a cryogenic superconducting Meissner shield and a four-layer μ -metal enclosure ($S_{\text{total}} > 10^9$).
4. *Detection Feasibility in High-Field Optical Lattices.* In a baseline 10 T fountain with half-second interrogation, the field phase shift is $\delta\phi_{\text{EM}} \approx 5.3 \times 10^{-10}\text{ rad}$. Operating at $B_0 = 40\text{ T}$ with a ten-second lattice hold boosts the signal by quadratic field and hold-time scaling to

$$\delta\phi_{\text{EM}} \approx 0.29\text{ }\mu\text{rad}. \quad (106)$$

At standard atom shot noise (10^7 atoms/s), this enables definitive 5σ detection of the field's gravitational curvature in 3.4 days of averaging, or under 9 hours with 10 dB spin-squeezed states.

VII. PHYSICAL LOCALITY IN FIELD RESPONSE

Configurations $(A_0, K_{1/6})$ and $(A_{1/2}, K_{1/6})$ on the coaxial cylinder have identically vanishing local curvature and are gauge equivalent on every contractible region surrounding the solenoid, yet their gauge-reduced Hessians possess distinct spectra, separated by the quantitative spectral gap $\Delta\Xi > 0.9731179060$ in their weighted heat traces.

This result fundamentally recasts the Aharonov–Bohm effect. Analyzing how fields warp spacetime and how different physical probes interact with geometry clarifies a sixty-five-year-old conceptual tension in gauge and gravitational physics.

A. Local field energy vs. test-particle kinematics

Historically, the Aharonov–Bohm effect has been interpreted through two primary tenets: (i) the electromagnetic potential A_μ is physically more fundamental than the local field strength $F_{\mu\nu}$; and (ii) holonomy is an exclusively gauge-theoretic property that strictly requires an electrically charged quantum matter wave (an electron) for its physical detection. This narrative is incomplete because it treats spacetime as an inert, unresponding background, and models the probe as an idealized test charge with no gravitational coupling.

In the real world, every physical particle has mass and energy, and must propagate along geodesics of the physical spacetime metric $\mathbf{g}_{\mu\nu}$. When an electron traverses the Aharonov–Bohm apparatus, its true relativistic action is $S = -m_e c \int ds + q \oint A$, accumulating both a gauge holonomy phase and a metric geodesic phase,

$$\Delta\phi_e = \Delta\phi_{\text{gauge}} + \Delta\phi_{\text{metric}} = -\frac{e}{\hbar}\Phi + \frac{m_e c^2}{\hbar}\Delta\tau. \quad (107)$$

The absence of observed metric back-reaction in early experiments stems from the electron’s charge-to-mass hierarchy, $\Delta\phi_{\text{gauge}}/\Delta\phi_{\text{metric}} \sim (e/m_e)/\sqrt{G} \sim 10^{35}$. In electron beam interferometry [3], the gauge phase is macroscopic ($\sim 10^3$ – 10^5 rad) while the metric geodesic shift is negligible ($\sim 10^{-32}$ rad). Conventional literature mistook this numerical dominance for an absolute selection rule, concluding that the Aharonov–Bohm effect is decoupled from spacetime geometry.

Any physical field possessing energy-momentum warps the spacetime metric around it. The field configuration is not merely an isolated connection A on a fixed manifold; it is a coupled geometric entity $\Psi = (A, K)$ whose shape tensor K encodes the extrinsic curvature and warping of spacetime. Around any closed cycle enclosing the field, the composite connection generates two commuting holonomies,

$$H^\oplus(\theta, k) = H^g(\theta) \oplus H^K(k) \oplus I_{\text{span}\{e_r, e_z\}}. \quad (108)$$

Neither holonomy determines the other, as θ varies independently of k . On contractible patches observers see zero curvature and gauge-trivial fields, but global topology is imprinted on both the gauge fiber and spatial metric.

B. Representation-Theoretic Probe Hierarchy

The apparent exclusivity of the traditional Aharonov–Bohm effect is an artifact of the probe historically employed. The three classes of physical probes form a natural representation-theoretic hierarchy.

The charged electron probe ($H^g \oplus H^K$, dominated by H^g).—An electron carries gauge charge $q \neq 0$ and mass $m_e > 0$. While its trajectory follows geodesics governed by $H^K(k)$, its internal phase is driven by $H^g(\theta)$. Because gauge coupling exceeds gravitational coupling by 10^{35} , the interference fringe shifts purely in response to $H^g(\theta)$, remaining practically blind to metric back-reaction.

The neutral geometric probe (H^K).—A photon or neutral atom carries zero electric charge and does not couple to A_μ . However, as demonstrated in Sections V and VI, neutral particles propagate along geodesics of $\mathbf{g}_k$. Their optical path, transit time, and matter-wave phase are determined by k and δ . Thus, neutral probes detect the field through its geometric holonomy $H^K(k)$, disproving the premise that neutral particles cannot detect an Aharonov–Bohm field.

The field’s own response ($H^g \otimes H^K$).—Unlike external particles that select a single block projection (H^g for electrons, H^K for neutral probes), the field’s own curvature-energy Hessian $L_{\theta,k}$ acts on fluctuations that couple spatial tangent vectors to internal fibers. These fluctuations transform in the *tensor-product representation* $H^g \otimes H^K$, where group characters multiply, $\chi_{\text{mixed}} = \chi_A(\theta)\chi_K(k)$:

$$\chi_{\text{mixed}} = \cos(2\pi m\theta) \cos(2\pi mk). \quad (109)$$

The regularized heat trace $\Xi_{\theta,k}(t)$ evaluates this joint character, reading both gauge holonomy and spacetime warping without requiring an external particle probe.

C. Theoretical Scope Boundaries

The theoretical boundaries of these results are defined by four specific structural properties.

Coaxial cylinder vs. general curved manifolds.—The quantitative spectral separation and tensor-product representation are established on the coaxial cylindrical solenoid geometry $M = [R_1, R_2] \times S^1 \times [0, L]$. Reconstructing nonflat same-loop gauge and spacetime holonomies from a single response spectrum on general curved manifolds remains an open problem.

Theoretical discrimination vs. laboratory detector.—The theorem establishes that $L_{\theta,k}$ retains global holonomy data as an intrinsic spectral property of the field equations. Translating this spectral discrimination into specialized laboratory detection with calibrated error bounds represents an active avenue of experimental design.

Singlet sector cancellation.—The spectral gap $\Delta\Xi > 0.9731179060$ is identical for the charged sector and the complete physical Hessian $L_{\theta,k}$, because the complementary singlet sector L_k^0 has zero dependence on the gauge parameter θ . The distinction is strictly preserved upon completion to the full physical state space.

Dynamical persistence.—When embedded as initial data in the Einstein–Yang–Mills system, $(A_0, K_{1/6})$ and $(A_{1/2}, K_{1/6})$ generate exact flat Lorentzian developments $(M_k, \mathbf{g}_k)$. Because the metric development is identical and flat for both configurations, local observers cannot distinguish them at any time τ , yet their comoving and fixed-time field-response Hessians remain separated throughout the spacetime.

D. Conclusion

The Aharonov–Bohm effect is not an isolated quantum gauge anomaly. It is the gauge projection of a joint field response in which fields warp spacetime, photon and matter-wave probes read geometry, charged probes read gauge phase, and the field’s own response intrinsically binds both. Because the field configuration intrinsically determines the response operator that reads it, the Aharonov–Bohm distinction resolves as a joint gauge-gravitational phenomenon.

TABLE II. The representation-theoretic hierarchy of probes in the coaxial cylindrical solenoid geometry. In physical spacetime, all matter waves propagate along metric geodesics (H^K); for charged electrons, minimal gauge coupling (H^g) dominates the metric phase by $(e/m_e)/\sqrt{G} \sim 10^{35}$, whereas neutral probes isolate the pure metric holonomy.

Probe Class	Physical System	Target Block	Group Character	Observable Readout
Charged matter	Electron (e^-)	$H^g \oplus H^K$ (H^g dominant)	$\cos(2\pi m\theta)$	Gauge Aharonov–Bohm phase
Neutral massless	Photon (γ)	$H^K(k)$ ($q=0$)	$\cos(2\pi mk)$	Optical path & transit time shift
Neutral massive	Cold atom (^{87}Rb , ^{88}Sr)	$H^K(k)$ (10^{21} boost)	$\cos(2\pi mk)$	Gravitational phase shift (2ω)
Coupled field	Curvature Hessian ($L_{\theta,k}$)	$H^g \otimes H^K$ (tensor product)	$\cos(2\pi m\theta)\cos(2\pi mk)$	Spectral gap in weighted heat trace

Appendix A: Spectral Decomposition of the Curvature Hessian

This appendix establishes the block diagonalization of the gauge-reduced curvature Hessian, derives the characteristic polynomial (51), computes the multiplicities of the nonzero mode spectrum (52), and evaluates the complementary singlet spectrum (64).

1. Charged Sector Block Reduction

Fix a J -eigenvalue σi on the complexified charged space $E_K = \text{span}\{e_2, e_3\}$ with eigenvector q_σ . Let $(t_1, t_2, t_3) = (e_\phi, e_r, e_z)$ denote the standard local orthonormal spatial frame for TM , where $t_1 = e_\phi$ aligns with the azimuthal circle around the solenoid, and let $(\omega^1, \omega^2, \omega^3) = (rd\phi, dr, dz)$ be its dual coframe. In the Lie algebra $\mathfrak{so}(TM \oplus N)$, define the generators

$$B = J_{t_1 e_1}, \quad M_\sigma = J_{e_1 q_\sigma}, \quad S_{\ell, \sigma} = J_{t_\ell q_\sigma} \quad (\ell = 1, 2, 3). \quad (\text{A1})$$

These generators satisfy the commutation relations

$$[B, M_\sigma] = -S_{1, \sigma}, \quad [B, S_{1, \sigma}] = M_\sigma, \quad (\text{A2})$$

while $[B, S_{2, \sigma}] = [B, S_{3, \sigma}] = 0$.

A charged tangent variation in the σ -eigenspace is represented by

$$a = \sum_{i=1}^3 a_i \omega^i M_\sigma, \quad (\text{A3})$$

$$\kappa = \sum_{i,j=1}^3 \kappa_{ij} \omega^i \odot \omega^j q_\sigma, \quad \kappa_{ij} = \kappa_{ji}. \quad (\text{A4})$$

The total bundle fluctuation one-form is

$$X_i = a_i M_\sigma + \sum_{\ell=1}^3 \kappa_{i\ell} S_{\ell, \sigma}. \quad (\text{A5})$$

For a Fourier mode $e^{in \cdot x}$ with covariant derivative $d_i = ip_i = i(n_i + \sigma \theta_i)$, the linearized bundle curvature components $\mathcal{L}_{\theta,k} v = D_{\nabla \oplus} X_v$ have twelve independent coefficients for $i < j$:

$$m_{ij} = d_i a_j - d_j a_i + k(\delta_{i1} \kappa_{j1} - \delta_{j1} \kappa_{i1}), \quad (\text{A6})$$

$$s_{ij}^1 = d_i \kappa_{j1} - d_j \kappa_{i1} - k(\delta_{i1} a_j - \delta_{j1} a_i), \quad (\text{A7})$$

$$s_{ij}^\ell = d_i \kappa_{j\ell} - d_j \kappa_{i\ell}, \quad (\ell = 2, 3). \quad (\text{A8})$$

Ordering the nine independent inputs as (a_1, a_2, a_3) and $(\kappa_{11}, \kappa_{12}, \kappa_{13}, \kappa_{22}, \kappa_{23}, \kappa_{33})$, the configuration-space norm and curvature Hessian quadratic form are

$$\|v\|_G^2 = 2 \sum_{i=1}^3 |a_i|^2 + \sum_{i=1}^3 |\kappa_{ii}|^2 + 2 \sum_{i < j} |\kappa_{ij}|^2, \quad (\text{A9})$$

$$H(v, v) = 2 \sum_{i < j} \left(|m_{ij}|^2 + \sum_{\ell=1}^3 |s_{ij}^\ell|^2 \right). \quad (\text{A10})$$

2. Physical Kernel Transversality

An infinitesimal internal gauge transformation $\xi M_\sigma \in \Gamma(\mathfrak{so}(N))$ acts by

$$a_i = -d_i \xi, \quad \kappa_{11} = k\xi, \quad \kappa_{ij} = 0 \quad ((i, j) \neq (1, 1)). \quad (\text{A11})$$

Direct substitution into Eqs. (A6)–(A8) verifies that $\mathcal{L}_{\theta,k} v_{\text{gauge}} = 0$. When $k \neq 0$, the gauge orbit is not purely in the connection coordinate; it possesses an essential component along κ_{11} .

In addition to this gauge-vertical direction, the kernel of $\mathcal{L}_{\theta,k}$ contains the non-gauge physical vector

$$w(d, k) = (-d_1(k^2 + d_1^2), -d_1^2 d_2, -d_1^2 d_3, 0, -kd_1 d_2, -kd_1 d_3, -kd_2^2, -kd_2 d_3, -kd_3^2). \quad (\text{A12})$$

Every coefficient in (A6)–(A8) vanishes on $w(d, k)$. Because its κ_{11} component is zero while $\kappa_{11} = k\xi \neq 0$ on the gauge orbit, $w(d, k)$ is transverse to the gauge orbit whenever $w(d, k) \neq 0$. If $w(d, k)$ vanishes identically, the vector $a_1 = 1, a_2 = a_3 = 0, \kappa = 0$ is annihilated. Thus, every spatial Fourier mode contains a non-gauge physical zero mode, establishing

$$\dim \ker L_{\theta,k}^{\text{ch}} = \infty. \quad (\text{A13})$$

3. Mode diagonalization on the range of the axial projector

Restricting to the range of the axial projector P_z isolates the modes with $k_z = 0$, where the covariant derivative along the azimuthal circle is

$$d_1 = ip = i(m + \sigma \theta), \quad d_2 = d_3 = 0. \quad (\text{A14})$$

The curvature components simplify to

$$(m_{12}, s_{12}^1) = (ipa_2 + k\kappa_{12}, -ka_2 + ip\kappa_{12}), \quad (\text{A15})$$

$$(m_{13}, s_{13}^1) = (ipa_3 + k\kappa_{13}, -ka_3 + ip\kappa_{13}), \quad (\text{A16})$$

$$s_{12}^2 = ip\kappa_{22}, \quad s_{12}^3 = ip\kappa_{23}, \quad (\text{A17})$$

$$s_{13}^2 = ip\kappa_{23}, \quad s_{13}^3 = ip\kappa_{33}, \quad (\text{A18})$$

while all components with indices $(2, 3)$ vanish. The variables a_1 and κ_{11} do not appear, corresponding to the two zero modes identified above.

For each pair (a_2, κ_{12}) and (a_3, κ_{13}) , taking into account the metric weights in (A9)–(A10), the operator acts via the 2×2 matrix

$$C(p, k) = \begin{pmatrix} ip & k \\ -k & ip \end{pmatrix}. \quad (\text{A19})$$

Its Gram matrix is

$$C(p, k)^\dagger C(p, k) = \begin{pmatrix} p^2 + k^2 & -2ipk \\ 2ipk & p^2 + k^2 \end{pmatrix}. \quad (\text{A20})$$

The characteristic polynomial of $C(p, k)^\dagger C(p, k)$ factors as

$$\det(\lambda I - C^\dagger C) = (\lambda - (p - k)^2)(\lambda - (p + k)^2) = 0, \quad (\text{A21})$$

yielding eigenvalues $\lambda_\pm = (p \mp k)^2$. Since this block occurs identically for (a_2, κ_{12}) and (a_3, κ_{13}) , both eigenvalues appear with multiplicity two.

The remaining shape variables $\kappa_{22}, \kappa_{23}, \kappa_{33}$ do not couple to the connection. Incorporating the combinatorial factors of 2 in the norm (A9) for off-diagonal tensors, the quadratic form on this sector evaluates to

$$H_{\text{shape}}(v, v) = 2p^2(|\kappa_{22}|^2 + 2|\kappa_{23}|^2 + |\kappa_{33}|^2) = 2p^2\|v\|_G^2, \quad (\text{A22})$$

yielding the threefold degenerate eigenvalue

$$\lambda_{\text{shape}} = 2p^2 \quad (\text{multiplicity } 3). \quad (\text{A23})$$

Multiplying the determinant factors yields the characteristic polynomial,

$$\begin{aligned} \det(\lambda I - G(p, k)) &= -\lambda^2(2p^2 - \lambda)^3 \\ &\quad \times ((p - k)^2 - \lambda)^2((p + k)^2 - \lambda)^2, \end{aligned} \quad (\text{A24})$$

establishing Eq. (51) and the mode multiplicities in Eq. (52).

4. Neutral singlet sector spectrum

A neutral singlet variation takes values in the abelian generator J and the line L_K ,

$$b = \sum_{i=1}^3 b_i \omega^i J, \quad (\text{A25})$$

$$h = \sum_{i,j=1}^3 h_{ij} \omega^i \odot \omega^j e_1, \quad h_{ij} = h_{ji}. \quad (\text{A26})$$

With generators $B_\ell = J_{t_\ell e_1}$, the bundle fluctuation is

$$X_i = b_i J + \sum_{\ell=1}^3 h_{i\ell} B_\ell. \quad (\text{A27})$$

On an angular-axial mode with covariant derivative d_i , the curvature coefficients are

$$u_{ij} = d_i b_j - d_j b_i, \quad (\text{A28})$$

$$r_{ij}^\ell = d_i h_{j\ell} - d_j h_{i\ell}, \quad (\ell = 1, 2, 3), \quad (\text{A29})$$

$$\tau_{ij}^\ell = k(\delta_{i1} h_{j\ell} - \delta_{j1} h_{i\ell}), \quad (\ell = 2, 3). \quad (\text{A30})$$

The background gauge holonomy θ does not appear because $D_A J = 0$ and $[A_\theta, J] = 0$, so θ acts trivially on the singlet sector.

On the range of P_z , setting $k_z = 0$, the variables b_1 and h_{11} do not enter the curvature; b_1 is gauge vertical for $m \neq 0$, and h_{11} is a physical zero mode representing the tangent to the family that deforms k . The non-vanishing curvature components decompose into two decoupled quadratic blocks:

$$H_{\text{conn}}^0(v, v) = 2m^2 \sum_{j=2}^3 (|b_j|^2 + |h_{1j}|^2), \quad (\text{A31})$$

$$H_{\text{shape}}^0(v, v) = 2(m^2 + k^2)(|h_{22}|^2 + 2|h_{23}|^2 + |h_{33}|^2). \quad (\text{A32})$$

Diagonalizing these two sectors yields the nonzero singlet spectrum

$$\text{Spec}_+(L_{m,k}^0) = \{m^2 \text{ (mult 4)}, 2(m^2 + k^2) \text{ (mult 3)}\}, \quad (\text{A33})$$

proving Eq. (64) and demonstrating that the singlet spectrum is strictly independent of the gauge holonomy θ .

Appendix B: Winding Enclosures for the Weighted Heat Trace

This appendix establishes the Poisson summation identity (55), verifies the coefficient domination inequality (59), computes the numerical bounding intervals (61)–(63), and establishes stability in the Banach space $\mathcal{W}_{1/2}$.

1. Poisson Winding Expansion

From Eq. (52), for a fixed J -eigenspace σ and covariant azimuthal momentum $p = m + \sigma\theta$, the sum over nonzero eigenvalues of $h_t(\lambda) = \lambda e^{-t\lambda}$ is

$$6p^2 e^{-2tp^2} + 2(p - k)^2 e^{-t(p-k)^2} + 2(p + k)^2 e^{-t(p+k)^2}. \quad (\text{B1})$$

Summing over $m \in \mathbb{Z}$ and $\sigma = \pm 1$ yields the total charged trace $\Xi_{\theta,k}^{\text{ch}}(t)$ in Eq. (53).

Under the standard Gaussian Poisson summation formula [40], for any $c, t > 0$ and $a \in \mathbb{R}$,

$$\sum_{m \in \mathbb{Z}} \exp(-tc(m+a)^2) = \sqrt{\frac{\pi}{ct}} \sum_{w \in \mathbb{Z}} \exp\left(-\frac{\pi^2 w^2}{ct}\right) \times \exp(2\pi i w a). \quad (\text{B2})$$

Both sides are smooth and rapidly decreasing in $t > 0$. Differentiating both sides with respect to $-t$ yields

$$\sum_{m \in \mathbb{Z}} c(m+a)^2 \exp(-tc(m+a)^2) = \sum_{w \in \mathbb{Z}} \mathcal{A}_c(w, t) \times \exp(2\pi i w a), \quad (\text{B3})$$

where the winding amplitude is

$$\mathcal{A}_c(w, t) = \sqrt{\frac{\pi}{c}} t^{-3/2} \left(\frac{1}{2} - \frac{\pi^2 w^2}{ct}\right) \exp\left(-\frac{\pi^2 w^2}{ct}\right). \quad (\text{B4})$$

Applying (B3) to each term of (53) decomposes the trace into pure gauge and hybrid sectors:

$$\sum_{m, \sigma} 3p^2 e^{-2tp^2} = 6 \sum_{w \in \mathbb{Z}} \mathcal{A}_2(w, t) \cos(2\pi w \theta), \quad (\text{B5})$$

$$\sum_{m, \sigma, \pm} (p \mp k)^2 e^{-t(p \mp k)^2} = 8 \sum_{w \in \mathbb{Z}} \mathcal{A}_1(w, t) \chi_{\text{mixed}}(w), \quad (\text{B6})$$

where $p = m + \sigma\theta$ and $\chi_{\text{mixed}}(w) = \cos(2\pi w \theta) \cos(2\pi w k)$. Summing all terms establishes the winding expansion (55). At $w = 0$, the winding amplitudes evaluate to $\mathcal{A}_c(0, t) = \frac{1}{2} \sqrt{\pi/c} t^{-3/2}$, yielding the zero-winding trace

$$\Xi^{(0)}(t) = 6\mathcal{A}_2(0, t) + 8\mathcal{A}_1(0, t) = \frac{\sqrt{\pi}}{2t^{3/2}} (3\sqrt{2} + 8), \quad (\text{B7})$$

which proves Eq. (56).

2. Proof of coefficient domination at $t = 1$

At $t = 1$, for any nonzero winding $m \neq 0$, set $x = \pi^2 m^2 \geq \pi^2 > 9.86$. Then

$$\mathcal{A}_1(m, 1) = \sqrt{\pi} \left(\frac{1}{2} - x\right) e^{-x}, \quad (\text{B8})$$

$$\mathcal{A}_2(m, 1) = \sqrt{\frac{\pi}{2}} \left(\frac{1}{2} - \frac{x}{2}\right) e^{-x/2}. \quad (\text{B9})$$

Both coefficients are strictly negative since $x > 1/2$. For $x \geq \pi^2 > 9.86$, the ratio of absolute values satisfies

$$\frac{|\mathcal{A}_1(m, 1)|}{|\mathcal{A}_2(m, 1)|} = 2\sqrt{2} \frac{x - 1/2}{x - 1} e^{-x/2} < 0.022 < \frac{3}{4}, \quad (\text{B10})$$

using the bounds $(x - 1/2)/(x - 1) < 1.06$ and $e^{-x/2} < 0.0073$. This proves Eq. (59). Consequently, for every shape amplitude $k \in \mathbb{R}$,

$$B_m(k; 1) \leq 6\mathcal{A}_2(m, 1) + 8|\mathcal{A}_1(m, 1)| < 0, \quad (\text{B11})$$

which strictly enforces $B_m(k; 1) < 0$ and establishes (60).

3. Certified numerical bounding intervals

To compute high-precision numerical values for $\Xi_{\theta, k}^{\text{ch}}(1)$ at $k = 1/6$, truncation of the lattice sum (53) at $|n| \leq N = 30$ and integration of the Gaussian tails for $c, t > 0$ yields

$$\sum_{|n| > N} (n+a)^2 e^{-tc(n+a)^2} \leq \frac{2(N-|a|)}{tc} e^{-tc(N-|a|)^2} + \frac{2}{tc} \int_{N-|a|}^{\infty} e^{-tcu^2} du. \quad (\text{B12})$$

For $N = 30$ and $t = 1$, the tail contribution is strictly bounded by 10^{-35} . Performing interval arithmetic on the finite lattice sum yields the bounding intervals

$$\begin{aligned} 10.363198088434 &< \Xi_{0, 1/6}^{\text{ch}}(1) < 10.363198088435, \\ 11.336315994496 &< \Xi_{1/2, 1/6}^{\text{ch}}(1) < 11.336315994497. \end{aligned} \quad (\text{B13})$$

Subtracting the outer endpoints proves

$$\Xi_{1/2, 1/6}^{\text{ch}}(1) - \Xi_{0, 1/6}^{\text{ch}}(1) > 0.9731179060. \quad (\text{B14})$$

Similarly, evaluating the derivative (68) at $\theta = k = 1/6$ and $t = 1$ gives

$$0.03739228008611 < \partial_k \Xi_{1/6, 1/6}^{\text{ch}}(1) < 0.03739228008612. \quad (\text{B15})$$

4. Stability on the weight space $\mathcal{W}_{1/2}$

Consider the Banach space

$$\mathcal{W}_{1/2} = \{f \in C([0, \infty)) : \|f\|_{1/2} = \sup_{x \geq 0} |f(x)| e^{x/2} < \infty\}. \quad (\text{B16})$$

For any $f \in \mathcal{W}_{1/2}$, the pointwise bound $|f(x)| \leq \|f\|_{1/2} e^{-x/2}$ holds. For the mode eigenvalues $\lambda \geq 0$,

$$\begin{aligned} |\text{Tr}(P_z f(L_{\theta, k}))| &\leq \|f\|_{1/2} \sum_{\sigma = \pm 1} \sum_{m \in \mathbb{Z}} \left[3e^{-p_{m, \sigma}^2} \right. \\ &\quad \left. + 2e^{-(p_{m, \sigma} - k)^2/2} + 2e^{-(p_{m, \sigma} + k)^2/2} \right]. \end{aligned} \quad (\text{B17})$$

The series converges uniformly to a finite constant $C_*(k)$. The difference functional

$$\mathcal{D}(f) = \text{Tr}(P_z f(L_{1/2, k})) - \text{Tr}(P_z f(L_{0, k})) \quad (\text{B18})$$

is linear and continuous with norm $\|\mathcal{D}\| \leq 2C_*(k)$. Since $\mathcal{D}(h_1) > 0.9731179060$, every function f satisfying

$$\|f - h_1\|_{1/2} < \frac{0.9731179060}{2C_*(k)} \implies \mathcal{D}(f) > 0. \quad (\text{B19})$$

Because the singlet spectrum is θ -independent, $\text{Tr}(P_z f(L_k^0))$ cancels identically in $\mathcal{D}(f)$ for all f . This establishes stability for the full physical Hessian on an open neighborhood in $\mathcal{W}_{1/2}$.

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